

Replisome loading reduces chromatin motion independent of DNA synthesis

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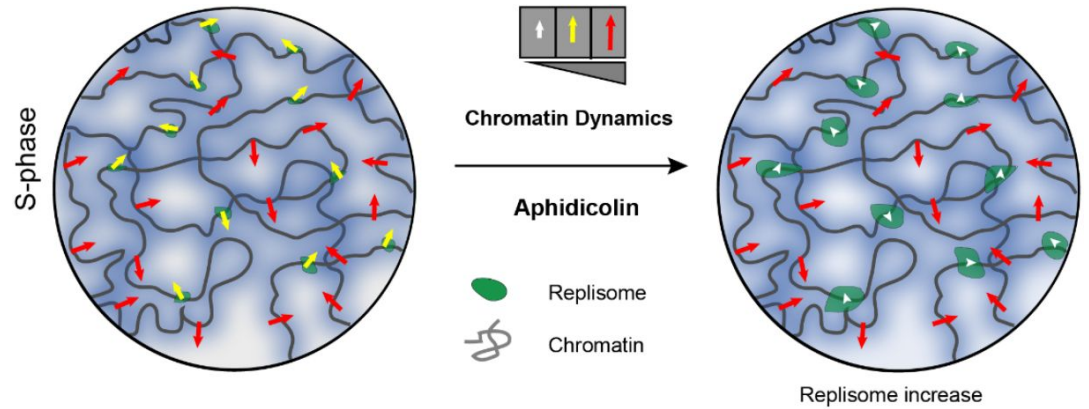
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Abstract

Summary

Chromatin has been shown to undergo diffusional motion, which is affected during gene transcription by RNA polymerase activity. However, the relationship between chromatin mobility and other genomic processes remains unclear. Hence, we set out to label the DNA directly in a sequence unbiased manner and followed labeled chromatin dynamics in interphase human cells expressing GFP-tagged PCNA, a cell cycle marker and core component of the DNA replication machinery. We detected decreased chromatin mobility during the S-phase compared to G1 and G2 phases in tumor as well as normal diploid cells using automated particle tracking. To gain insight into the dynamical organization of the genome during DNA replication, we determined labeled chromatin domain sizes and analyzed their motion in replicating cells. By correlating chromatin mobility proximal to the active sites of DNA synthesis, we showed that chromatin motion was locally constrained at the sites of DNA replication. Furthermore, inhibiting DNA synthesis led to increased loading of DNA polymerases. This was accompanied by accumulation of the single-stranded DNA binding protein on the chromatin and activation of DNA helicases further restricting local chromatin motion. We, therefore, propose that it is the loading of replisomes but not their catalytic activity that reduces the dynamics of replicating chromatin segments in the S-phase as well as their accessibility and probability of interactions with other genomic regions.



Highlights

- - Direct and sequence unbiased labeling of DNA genome-wide
- - DNA labeled chromatin is more mobile in G1/G2 relative to the S-phase
- - Restriction of chromatin motion occurs proximal to sites of DNA replication
- - Loading of replisomes, even in the absence of processive DNA synthesis, restricts chromatin motion

eLife assessment

This is a **valuable** investigation of the chromatin dynamics throughout the cell cycle by using fluorescence signals and patterns of GFP-PCNA and CY3-dUTP, which labels newly synthesized DNA. The authors report reduced chromatin mobility in S relative to G1 phase. The technology and methods used are **solid**. The data will be of interest to researchers working on chromatin dynamics.

Introduction

Dynamic yet functionally stable organization of cellular processes is a crucial feature of biological systems, which allows them to respond to external stimuli and survive. The eukaryotic nucleus is a complex subcellular organelle where DNA metabolism including its replication, repair and transcription, occurs. Eukaryotic DNA is organized in the nuclear space by interactions with histones and architectural proteins to form a hierarchy of domains and compartments of the interphase chromatin. Nuclear architecture is dynamically modulated due to the binding of biomolecules and epigenetic changes of the chromatin. It is also interdependent with DNA metabolism mediated by the action of enzymes on the chromatin. The maintenance of the DNA (including its replication and repair) and its transcription into RNA are spatio-temporally organized within the cell nucleus.

Analysis of the local chromatin dynamics in live cells revealed that an essential aspect of interphase chromatin is its mobile nature (Gasser, 2002; Marshall et al., 1997). The movement of chromatin loci was shown to be consistent with an anomalous (constrained) diffusion model (Scipioni et al., 2018; Shukron et al., 2019). This model indicates that a single chromatin locus is corralled within a sub-micron radius and exhibits random diffusion motion and will execute multiple random jumps into neighboring compartments (Bronshtein et al., 2016; Chubb et al., 2002; Heun et al., 2001; Levi et al., 2005; Marshall et al., 1997). This behavior, which we refer to as local chromatin diffusion (LCD), has been described in multiple systems, suggesting that it is likely to represent a fundamental aspect of chromatin dynamics in eukaryotes.

According to the current paradigm, the 4D organization of the chromatin inherently includes its physical properties as a long polymer (Esposito et al., 2021, 2019), while stochastic thermodynamically driven events are likely to play a key role in the domain organization of the chromatin (Conte et al., 2020; Shin and Brangwynne, 2017) and in the regulation of genomic processes (Hnisz et al., 2017; Kilic et al., 2019; Laghmach et al., 2021; Nozaki et al., 2017; Spegg and Altmeyer, 2021; Uchino et al., 2022).

Some studies have reported that chromatin mobility is enhanced due to active transcription (Gu et al., 2018; Tunnacliffe and Chubb, 2020), whereas others report rather a decrease in mobility (Mach et al., 2022). Furthermore, other studies report diverse effects of RNA polymerase II inhibition on chromatin motion (Germier et al., 2017; Ku et al., 2022; Shaban et al., 2018). It has been also shown that the removal of RNA polymerase II from chromatin relaxes chromatin and increases its mobility (Babokhov et al., 2020). Conversely, there is an established view that chromatin mobility at the sites of double-strand DNA breaks increases concomitant with their repair (Eaton and Zidovska, 2020; Hauer and Gasser, 2017; Hauer et al., 2017; Nagai et al., 2010). Analysis of fluorescently tagged histones using displacement correlation spectroscopy has shown that chromatin undergoes coherent micron-scale motion at the time scales of 5-10 seconds independently of the cell cycle stage in mammalian cells (Zidovska et al., 2013). This coherent motion extended beyond individual chromosomes, suggesting mechanical coupling between chromosomes. Furthermore, the correlated motion of chromatin was ATP-dependent and completely disappeared upon DNA damage induction (Eaton and Zidovska, 2020; Zidovska et al., 2013).

DNA replication is a highly conserved energy-dependent process occurring in S-phase of the cell cycle, when chromatin structures undergo extensive reorganization to facilitate DNA synthesis (Vincent et al., 2008). An early study in budding yeast (Heun et al., 2001) demonstrated that individual heterologous loci became constrained in S-phase when integrated close to early- and late-firing replication origins, but not at the telomeric or centromeric regions. However, changes in chromatin mobility in S-phase were not observed when analyzing it at the level of chromosome territories in mammalian cells (Walter et al., 2003). Recent work using a CRISPR-based DNA imaging system suggests that local chromatin motion is restricted upon S-phase entry and more markedly in mid-late S-phase (Ma et al., 2019).

Altogether, it is not clear whether and how chromatin mobility changes during DNA replication and a mechanism behind the changes in chromatin motion. Therefore, it is important to address how changes in structure and metabolism of chromatin affect its mobility. It is quite intriguing to postulate that the process of genome duplication in mammals, which is performed at the level of naked DNA and involves local chromatin decondensation and rearrangements at the complete hierarchy of domains (Baddeley et al., 2010; Chagin et al., 2019; Löb et al., 2016; Sadoni et al., 2004; Sporbert et al., 2002) is associated with changes in chromatin mobility. Furthermore, it is tempting to speculate that the modulation of LCD may play a regulatory role; for example, by helping to define the transcriptional profile of the nucleus, by provoking collisions between regulatory regions, promoter regions and transcription factories. These events could be halted or slowed down during the replication of the genome, avoiding collisions of the

transcription with the replication machineries. An alternative but not mutually exclusive model is that changes in LCD result from the execution of nuclear processes such as transcription or replication. This is particularly appealing as DNA/RNA helicases and polymerases are, in essence, motor proteins that reel DNA through. To distinguish between these possibilities, alterations in LCD have to be characterized within the context of relevant nuclear processes and by labeling DNA directly and in an unbiased manner.

The process of genome replication has a particular and intrinsic connection between chromatin organization and the spatio-temporal progression of genome replication (reviewed in (Mamberti and Cardoso, 2020 [↗](#))). In that sense, firing of origins of replication by the activation of DNA helicase complexes followed by the loading of synthetic polymerase complexes tracks chromatin compaction and upon DNA duplication the focal chromatin organization at multiple hierarchical levels is preserved and can be detected over several cell generations (Cremer et al., 2020 [↗](#); Jackson and Pombo, 1998 [↗](#); Sadoni et al., 2004 [↗](#); Sparvoli et al., 1994 [↗](#)). Importantly, genome replication is the only DNA metabolic process that encompasses the entire genome, thus ensuring the preservation of the genetic material upon cell division.

As most of the studies introduce artificial DNA sequences in genomic loci and use a large array of chromatin binding proteins to visualize the loci, chromatin dynamics may be altered in the subsequent process (Germier et al., 2017 [↗](#)). Therefore, a more direct way to measure chromatin dynamics is to label and track the DNA directly (Schermelleh et al., 2001 [↗](#)). A similar procedure has previously been used to mark chromosome territories and characterize their long-term rearrangements (Bornfleth et al., 1999 [↗](#); Pliss et al., 2009 [↗](#); Walter et al., 2003 [↗](#)).

In this study, we investigated the mobility of chromatin in human cells, focusing on how changes in chromatin mobility are influenced by cell cycle progression and, in particular, DNA replication. To achieve this, we performed a detailed analysis of chromatin mobility in S-phase by combining locus-independent global labeling of DNA with reliable particle tracking. Measurement of the DNA content of the labeled structures allowed us to elucidate whether DNA replication affects chromatin mobility at the level of replication domains. Our results show that chromatin mobility generally decreases during S-phase and, in particular, at the proximity of the DNA polymerase complexes. Furthermore, we extended our study to dissect mechanisms behind the S-phase related changes in chromatin mobility and inhibited DNA synthesis using small molecule inhibitors. We showed that chromatin mobility is further decreased in S phase after inhibition of DNA synthesis. These results imply that loading of the polymerase complexes rather than the synthesis of DNA *per se* restraints DNA mobility.

Results and discussion

Genome-wide labeling of DNA and quantification of labeled chromatin domains

To evaluate LCD (local chromatin diffusion) relative to the cell cycle stage, we first developed an experimental system to monitor both replication and chromatin changes in living cells in real-time. We generated HeLa cell lines that stably express GFP-tagged proliferating cell nuclear antigen (PCNA) and single-stranded DNA binding protein (RPA) (Methods, **Supplementary Table 1** [↗](#)). We transfected fluorescent PCNA plasmid to label replication sites in human normal diploid fibroblasts (IMR90) (Nichols et al., 1977 [↗](#)). PCNA is a core component of the DNA replication machinery and a marker for cell cycle progression (**Figure 1A** [↗](#)) (Chagin et al., 2016 [↗](#); Easwaran et al., 2005 [↗](#); Leonhardt et al., 2000 [↗](#); Moldovan et al., 2007 [↗](#); Prelich et al., 1987 [↗](#)). To visualize the mobility of native chromatin, we took the advantage of the ongoing DNA replication. We delivered a pulse of the fluorescently labeled nucleotide Cy3-dUTP by electroporation into an asynchronously growing population of human HeLa GFP-PCNA tumor cells and human diploid

IMR90 fibroblasts, which allowed us to study chromatin dynamics in a global genome-wide manner (Methods, **Figure 1A**). The nucleotide is incorporated into the nascent DNA of the cells in various periods of S-phase, effectively labeling the chromatin directly in an unbiased manner (Manders et al., 1999; Sadoni et al., 2004; Schermelleh et al., 2001).

The Cy3-dUTP labeled chromatin structures were stable over the cell cycle progression and in subsequent cell cycles. Using time lapse microscopy, we followed the cells that incorporated nucleotides in the initial S phase stage over subsequent cell cycles. We used GFP-PCNA nuclear pattern to determine the cell cycle stages and sub-periods of S-phase (Methods, Microscopy). This allowed us to classify cells in different cell cycle stages and sub-periods of S-phase (G1, early S, mid S, late S, G2), which is illustrated in **Figure 1B** (see also movies 1-5). With this approach, DNA labeled during the pulse of Cy3-dUTP nucleotide corresponds to genomic regions replicated concomitantly during an S-phase sub-period. Since LCD measurements depend on the object size, it was important to evaluate the size of the labeled DNA domains. This allowed us to correlate the chromatin domain sizes and their diffusion rates. For this purpose, we measured the total DNA amount in a cell and the fraction of it that corresponded to the labeled domain (Methods, DNA quantification). First, we applied chemical fixation to cells labeled with Cy3-dUTP using formaldehyde. The total DNA was then labeled using the DNA dye DAPI. Next, we segmented the entire nucleus as well as the individual labeled chromatin foci within the same cell. The fraction of DAPI intensity within the segmented replication focus (I_{RF}) over the total DNA intensity within the cell ($I_{DNA\ total}$) yields the amount of DNA present per labeled chromatin focus (**Figure 2A**, **Supplementary Figure S1**). Since nuclear DNA amount doubles continuously throughout the S-phase (Chagin et al., 2016; Leonhardt et al., 2000), (**Figure 2B**) it was important to scale the total DNA amount by a correction factor depending on S-phase sub-stage to measure the DNA amount per focus more accurately. The relative amount of DNA throughout the cell cycle stages and sub-stages of S-phase was calculated and plotted as histograms, with the mean of the histogram for each cell cycle (sub)stage constituting the cell cycle correction factor (**Figure 2B**). The fraction of DAPI intensities were corrected by multiplication with the genome size corresponding to the cell cycle stage. The genome size of HeLa Kyoto cells is $GS = 9.682 \pm 0.002$ Gbp (Chagin et al., 2016) and for IMR90 fibroblasts the genome size is 6.37 Gbp as measured earlier (Nichols et al. 1977). We plotted the DNA amount present in each replication focus in **Figure 2C** for HeLa on left and IMR90 on right. The highest frequency of average DNA amount per focus (mode + 1 bin) was about 300-600 kbp of DNA (**Figure 2C**). Altogether, with our labeling approach, we labeled DNA domains of sizes ranging from 0.5 Mbp to 10 Mbp, with the vast majority corresponding to 0.5 Mbp, which correspond well to multi-loop chromatin domains corresponding in size to topological associated domains (TADs) (reviewed in (Giorgetti and Heard, 2016)).

Chromatin motion decreases in the S-phase of the cell cycle relative to the G1 and G2 phases

To determine how the global dynamics of chromatin changes during cell cycle progression, we used LCD measurements relative to the cell cycle stage. Live cell time-lapse image sequences of HeLa and IMR90 cells after labeling chromatin with Cy3-dUTP were obtained and motion analysis was performed to determine the type of motion (**Figure 3A**, Methods). Normal diffusion or Brownian motion is a linear diffusion model with $\alpha = 1$ and when $\alpha > 1$ it is termed super diffusion. First, the cells were annotated according to the different cell cycle stages (G1, S, G2) based on the PCNA subnuclear pattern (Methods). PCNA forms puncta or foci at the active replication sites during S-phase and this was used to classify cells in S-phase. We were able to distinguish between G1 and G2 cells, even though they exhibit a similar diffused PCNA subnuclear distribution, based on the information on the preceding cell cycle stage from the time lapse analysis performed after Cy3-dUTP labeling (Methods, Microscopy). Specifically, cells with diffusely distributed PCNA signal which had previously undergone mitosis were in G1 phase,

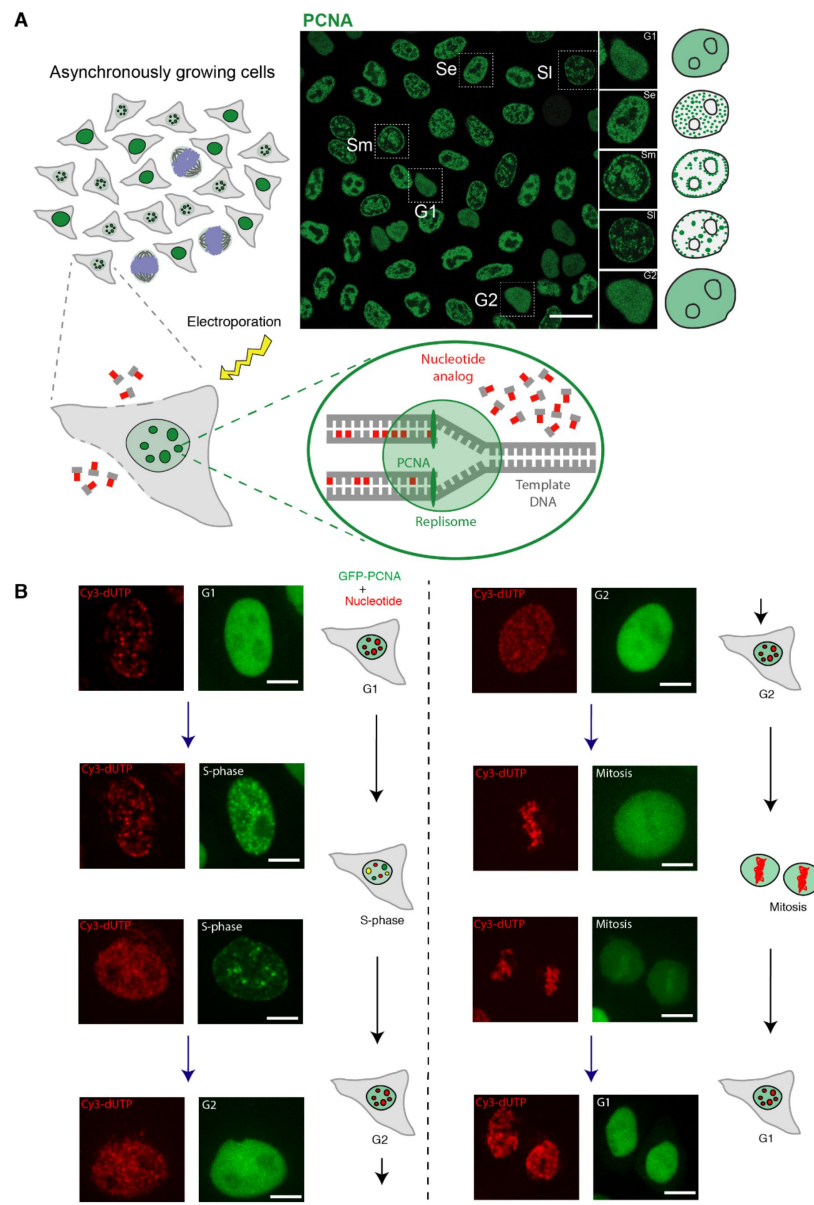


Figure 1

Incorporation of Cy3-dUTP in HeLa cell nuclei labels the whole genome randomly and with equal probability.

(A) Schematic illustration of the labeling system for monitoring chromatin mobility and cell cycle progression. During S-phase, PCNA accumulates within the nucleus at sites of active DNA replication and exhibits a distinct puncta pattern. During G1 and G2, GFP-PCNA is diffusely distributed throughout the nucleus. Asynchronously growing populations of cells were exposed to electroporation to promote the uptake of Cy3-dUTP. In cells undergoing DNA replication, this fluorescent nucleotide is incorporated into nascent DNA strands at sites of active DNA replication, resulting in the direct fluorescent labeling of genomic segments. Based on the PCNA pattern, different cell cycle stages can be differentiated as shown in the image on the right (Se-Early S, Sm-Mid S, SL-Late S, G1/G2-Gap phases, Green - PCNA). **(B)** After Cy3-dUTP labeling (shown in red), cells were followed by time lapse microscopy to identify the cell cycle (sub)stages and their progression. The representative images of different cells using time lapse microscopy were shown to depict the patterns of PCNA (shown in green) in each substage and their change over time (colocalized signals in yellow). This was used to classify cells in G1, S and G2 phases of the cell cycle for motion analysis. Approximately 18 to 24 hours after nucleotide electroporation, Cy3-dUTP-labeled cells were imaged for motion analysis (see also movies 1-5). The contrast of the images was adjusted linearly for visualization purposes. Scale bar: 5 μ m.

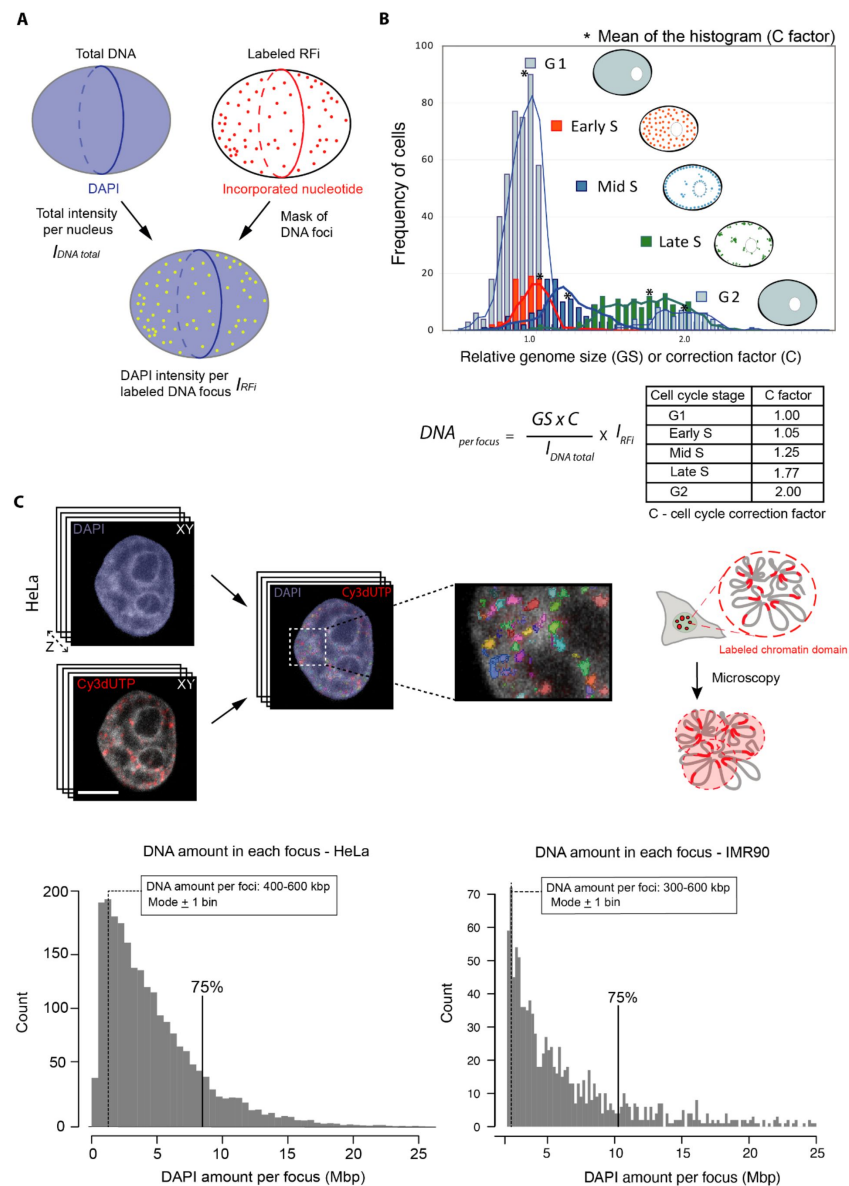


Figure 2

Principle of measuring DNA content per labeled DNA focus using confocal data.

(A) To determine the amount of DNA per labeled DNA focus; we used the total DAPI signal (DNA amount) of the segmented whole nucleus ($I_{DNA\ TOTAL}$). The DNA intensity per labeled DNA focus within the segmented foci is obtained (I_{RFi}) by masking the replication foci and estimating the corresponding portion of DAPI signal. **(B)** Throughout the S-phase progression the amount of DNA increases twofold from early to late S-phase. The amount of DNA present in the nucleus at a particular cell cycle stage can be determined by measuring the DNA amount in the population of cells, while using the PCNA pattern to determine the cell cycle stage and S-phase sub-stage (see also [Figure 1](#)). The relative mean amount of DNA of each of the cell cycle (sub-)stages is used to calculate the cell cycle correction factor. The cell cycle correction factor (C/cell cycle stage) was estimated as: 1.0/G1; 1.05/early S-phase; 1.25/mid-S-phase; 1.77/late S-phase, 2/G2. The G1 genome size (GS) for HeLa cells is 9.7 Gbp ([Chagin et al., 2016](#)). The amount of DNA per labeled focus is the ratio of I_{RFi} and $I_{DNA\ TOTAL}$ multiplied by C x GS. **(C)** The illustration on right depicts the imaging of labeled replication foci using confocal microscopy. DNA quantification of replication labeled foci in tumor HeLa and normal diploid IMR90 cells was done by imaging full z-stacks volume of chromatin labeled with Cy3-dUTP and DNA with DAPI and imaged using confocal spinning disk microscopy ([Supplementary Figure S1](#), [Supplementary Table 5](#)). The histogram represents the DNA amount per focus for labeled S-phase cells (N= 30 cells) for HeLa and IMR90 cells. The mode $\pm$ 1 bin of the histogram represents the highest frequency of average size of replication domains labeled (300-600 kbp). Scale bar: 5 μ m.

whereas the ones with similar diffuse PCNA pattern that had previously undergone S-phase (punctated PCNA pattern) were classified as being in G2 phase (**Figure 1B**). The PCNA signal was also used to segment the nucleus, and the individual chromatin foci were detected within the segmented nuclei. Probabilistic tracking was performed to obtain individual chromatin trajectories (**Figure 3B**; **Supplementary Figure S2**). In case of IMR90 cells, affine image registration was performed using the method in (Celikay et al., 2022) to address the stronger cell movement compared to HeLa cells. This was followed by a Mean Square Displacement (MSD) analysis to determine the chromatin motion in different cells (**Supplementary Figure S2**). In fixed cells, labeled chromatin foci showed almost no motion, which was used as a control for the stability of the imaging system and the tracking protocol. We plotted the MSD over time (up to 20 s) for chromatin foci in cells from different cell cycle stages as well as for fixed cells (**Figure 3C**). As we focussed on chromatin mobility changes during S-phase, the G1, G2 cells were together in **Figure 3C**. The mean square displacement curves of G1, G2, S-phase (separated) are plotted in **Supplementary Figure 2B**.

We observed significantly constrained global chromatin motion in S-phase cells compared to non-replicating G1/G2 cells suggesting that chromatin was more constrained during DNA replication. This effect was stronger in IMR90 cells compared to HeLa Kyoto. The table shows the average diffusion rates (**Figure 3D**). For HeLa average diffusion rate of chromatin in G1/G2 was $D = 133.6 \mu\text{m}^2/\text{s} \times 10^{-5}$, whereas the diffusion rates dropped to $D = 105.6 \mu\text{m}^2/\text{s} \times 10^{-5}$ during S-phase (**Figure 3C**). For IMR90 average diffusion rate of chromatin in G1/G2 was $D = 86.5 \mu\text{m}^2/\text{s} \times 10^{-5}$, whereas the diffusion rates dropped to $D = 43.7 \mu\text{m}^2/\text{s} \times 10^{-5}$ during S-phase (**Figure 3C**). We computed the α values in different stages, which define the type of diffusion motion. Chromatin exhibited anomalous subdiffusion or obstructed diffusion with $0.1 < \alpha < 0.9$. Anomalous diffusion of cellular structures including chromatin with α values between 0.1 and 0.9 have been reported (Bronshtein et al., 2015; Ghosh and Webb, 1994; Mach et al., 2022; Oliveira et al., 2021; Simson et al., 1998; Smith et al., 1999).

In agreement with our results, it has been initially reported in yeast that some chromatin loci are constrained during S-phase (Heun et al., 2001). This study has been extended to the mammalian genome using the CRISPR targeted labeling of specific genomic loci to demonstrate that the S-phase mobility of the labeled chromosomal loci decreases in S-phase compared to G1/G2 (Ma et al., 2019). Another study reported that during DNA replication there were changes in chromatin mobility due to an unknown mechanism (Nozaki et al., 2017).

As we measured decrease in global chromatin motion during S-phase, which includes labeled chromatin which is replicating as well as non-replicating, we next focused the study on the microenvironment of active replication sites. This opened the question of whether the loading of the replisome on chromatin or its enzymatic activity during S-phase actively restricted chromatin motion. Hence, we analyzed in detail the spatial relationship of chromatin diffusion and DNA replication sites.

Chromatin motion decreases in proximity to active DNA replication sites

DNA replication involves systematic and structured assembly of proteins directly or indirectly involved in DNA synthesis. DNA replication factors such as DNA polymerase clamp protein (PCNA), the DNA helicase complex that unwinds DNA, and the single-stranded DNA binding protein A (RPA) complex, which stabilizes and protects the single stranded DNA exposed upon helicase activity are illustrated in (**Figure 4A**). The DNA polymerase clamp PCNA, one of the most well studied replication proteins, was used to mark the active DNA replication sites. To test whether DNA replication factors restrict chromatin motion, we performed proximity analysis (Methods, **Figure 4B**). As before, we used Cy3-dUTP to label chromatin in the S-phase of the previous cell cycle. We then followed the cells through the cell cycle to select cells in which some

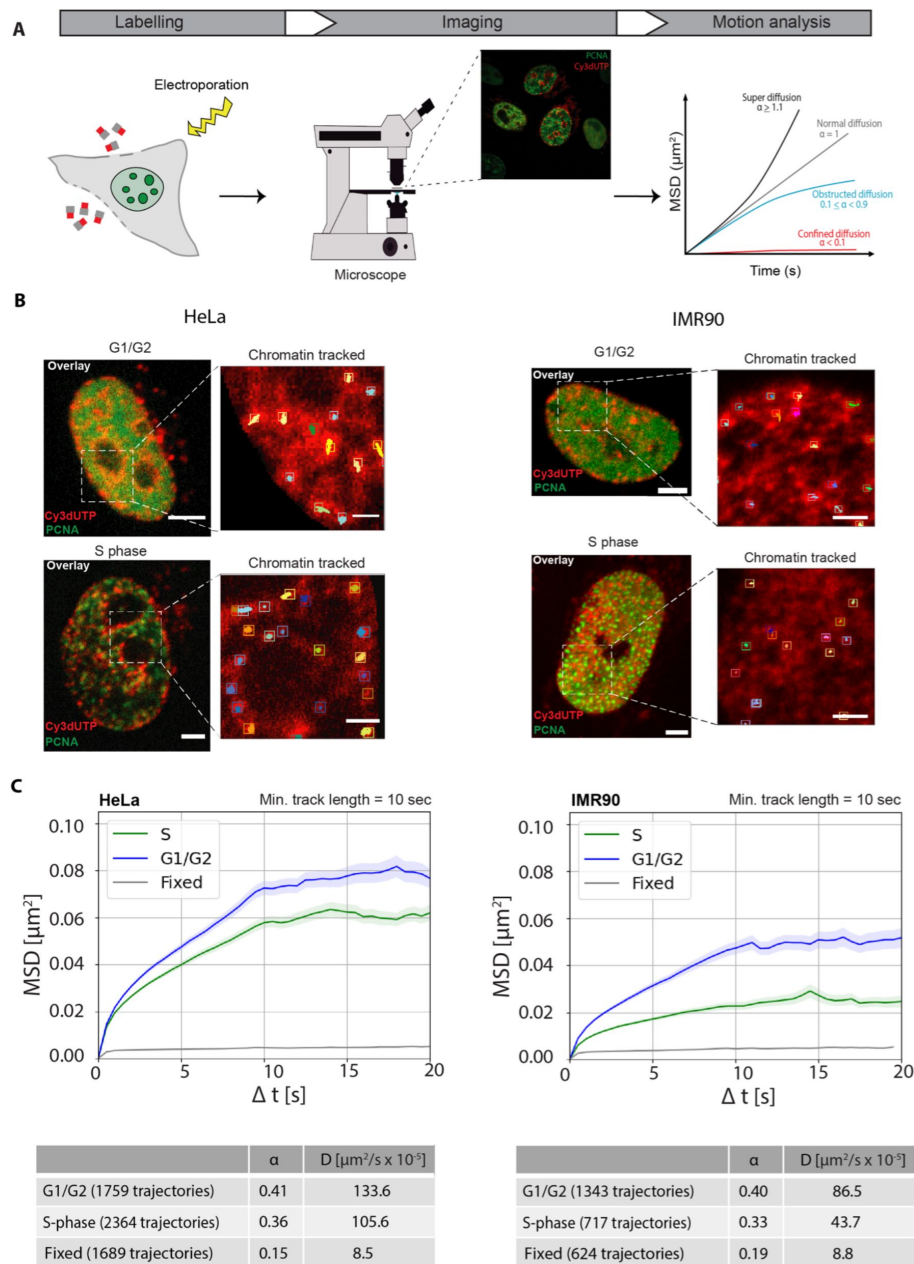


Figure 3

Single-particle motion analysis of labeled chromatin throughout the cell cycle.

(A) Brief schematics of the main steps of motion analysis starting with chromatin labeling using fluorescently labeled nucleotides via electroporation, followed the next day (i.e., cell cycle) by confocal time-lapse imaging of the chromatin channel and performing motion analysis on computed tracks to determine the diffusion rates of chromatin (Methods). **(B)** Overlay images of HeLa Kyoto and IMR90 cells expressing GFP/miRFP-PCNA and labeled chromatin (Cy3dUTP) in different cell cycle stages (G1/G2 – diffused PCNA, S-phase – PCNA puncta). Cropped region (white box) showing the chromatin tracks of individual foci in both G1/G2 and S-phase cells. The aggregates of Cy3-dUTP that are found in cytoplasm are excluded from the analysis using a nuclear mask. See also movies 1-5. **(C)** Result of motion analysis of computed chromatin tracks for different cell cycle stages (**Supplementary Figure S2**) for HeLa and IMR90 cells. Mean Square Displacement (MSD, μm^2) curves were plotted over time (s). Mean square displacement curves for G1/G2, S-phase, fixed cells with a minimum track length of 10 seconds and a total time of 20 seconds were plotted with error bars (SEM - standard error of the mean) representing the deviations between the MSD curves for an image sequence in transparent color around the curve. Scale bar: 5 μm . Insets scale bar: 1 μm .

of the sites of labeled chromatin were replicating in the S-phase of the next cell cycle at the time of observation. This allowed us to image the labeled chromatin marked in the previous cell cycle together with a live-cell marker (fluorescent PCNA) for the active replication sites in the next cell cycle (**Figure 4B**). Subsequently, we measured the mobility of chromatin from these S-phase cells at increasing center to center distances (CCD, R) from active replication sites (**Figure 4C**). For chromatin outside the CCD with replication sites in these S-phase cells, we observed the same diffusion rate as before for the chromatin foci in S-phase cells with no differentiation of whether chromatin was actively replicating or not (**Figure 3**). However, we observed that the chromatin in the proximity of replication sites (actively replicating) had more restricted motion when located up to 1 μm (center to center) distance to an active replisome, and this effect vanished at higher distances (**Figure 4C**, **Supplementary Figure S3**). These data indicate that the reduction of chromatin motion in S-phase is spatially correlated with DNA replication and suggest that DNA synthesis restricts chromatin motion. Hence, we next investigated whether loading of the DNA replication machinery restricts chromatin motion or alternatively DNA synthesis activity is responsible for it.

DNA synthesis inhibition leads to activation of DNA helicases and accumulation of single stranded DNA binding proteins and DNA polymerases

During DNA replication, replisome components are assembled at the origin of replication to form an active replisome (Casas-Delucchi and Cardoso, 2011; Yao and O'Donnell, 2010, 2016). To test whether the process of DNA synthesis itself is responsible for constraining chromatin, we analyzed chromatin motion after inducing replication stress. By treating cells with aphidicolin, DNA synthesis is slowed down or stopped altogether (Vesela et al., 2017). Aphidicolin is a tetracyclic antibiotic isolated from *Nigrospora sphaerica*, which interferes with DNA replication directly by inhibiting DNA polymerases α , ϵ , and δ (Bambara and Jessee, 1991; Baranovskiy et al., 2014; Byrnes, 1984; Cheng and Kuchta, 1993). Our hypothesis was that it is the loading of replisome components that affects the chromatin motion (LCD). Therefore, we focused on LCD measurements after inhibiting DNA synthesis directly with aphidicolin and characterized the effects on chromatin motion in order to understand the mechanism behind it.

First, we tested in detail the rate and level of inhibition of DNA synthesis with aphidicolin (150 μM) using thymidine analogs (in this case EdU), which get incorporated into newly synthesized DNA and can be detected using click-IT chemistry (Methods). We visualized GFP-PCNA and EdU in fixed cells and performed high-throughput image analysis to characterize the effect of aphidicolin on DNA synthesis inhibition at different timepoints (Methods, **Supplementary Figure S4**). We observed that DNA synthesis was inhibited minutes after aphidicolin treatment. Using high-content microscopy, we quantified the population of cells actively synthesizing DNA (EdU signal) upon stress and observed that in almost 99 % of the cell population, DNA replication was inhibited within half an hour of aphidicolin incubation (Methods, **Figure 5A**, **Supplementary Figure S4**, **S5**). Subsequent experiments were all performed with these conditions.

Secondly, we made use of the above conditions in which DNA synthesis was inhibited, and analyzed the consequences of replication stress on the replisome components and their kinetics. For this purpose, we performed time-lapse microscopy of GFP-PCNA and GFP-RPA34 expressing cells. During active DNA synthesis, the DNA polymerase clamp and processivity factor PCNA is loaded onto the DNA as a trimeric ring and is tightly bound to the DNA (**Figure 5A**). During aphidicolin treatment though, PCNA dissociated from DNA as shown before (Görisch et al., 2008) (Rausch et al., 2021) (**Figure 5A**). Aphidicolin treatment does not stop helicase activity and the single stranded DNA binding protein RPA is loaded on the ssDNA after being unwound by the DNA helicase. The more the DNA double helix is unwound, the more RPA loads onto the ssDNA generated (Rausch et al., 2021). For this analysis, we generated a HeLa cell line stably expressing

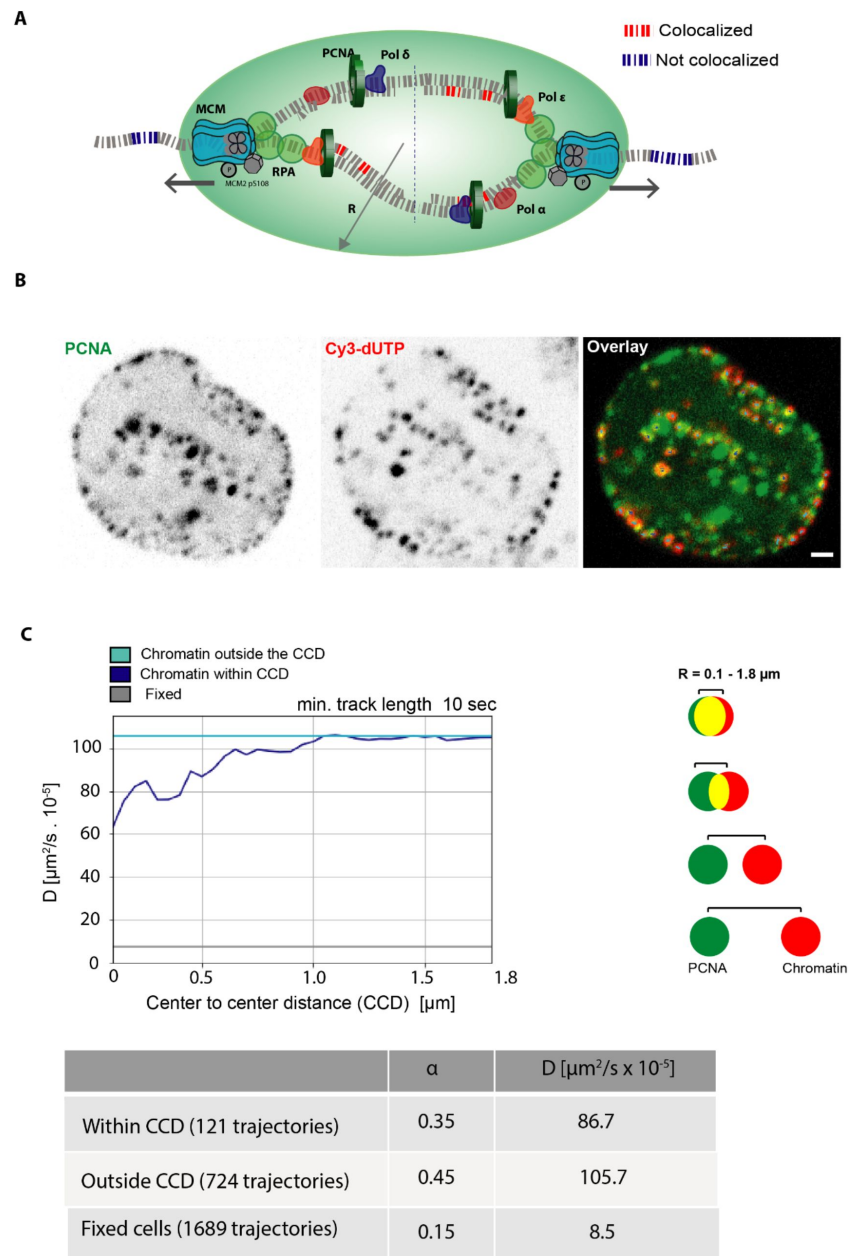


Figure 4

Analysis of chromatin mobility versus distance (proximity) to the DNA replication machinery.

(A) Schematic illustration of replisome components (helicase, replication protein A, proliferating cell nuclear antigen) actively replicating chromatin. The geometric centers of the labeled chromatin foci and labeled replication sites were first defined. The chromatin within the defined center to center distance (CCD) to a PCNA labeled replication site is defined as chromatin that is within CCD and, otherwise, is defined as outside the CCD. **(B)** In order to obtain mobility information of labeled chromatin in the proximity of PCNA foci (active replication sites) one frame of PCNA channel was acquired followed by 50 frames of the chromatin channel with a frame rate of 0.5 seconds. The images show the spatial distribution of PCNA and chromatin foci (Cy3-dUTP). **(C)** The graph represents the average diffusion rates of the mean square displacement curves (MSD) of chromatin within the center to center distance (CCD) and chromatin outside the center to center distance (CCD) with increasing distance (R) measured between the centers of PCNA and chromatin foci ([Supplementary Figure S3](#)). The table below provides the detailed information on number of trajectories per individual sample along with average diffusion rates ($\mu\text{m}^2/\text{s} \times 10^{-5}$) and anomalous α coefficient showing subdiffusion at 0.5 μm center to center distance (CCD). Scale bar: 1 μm .

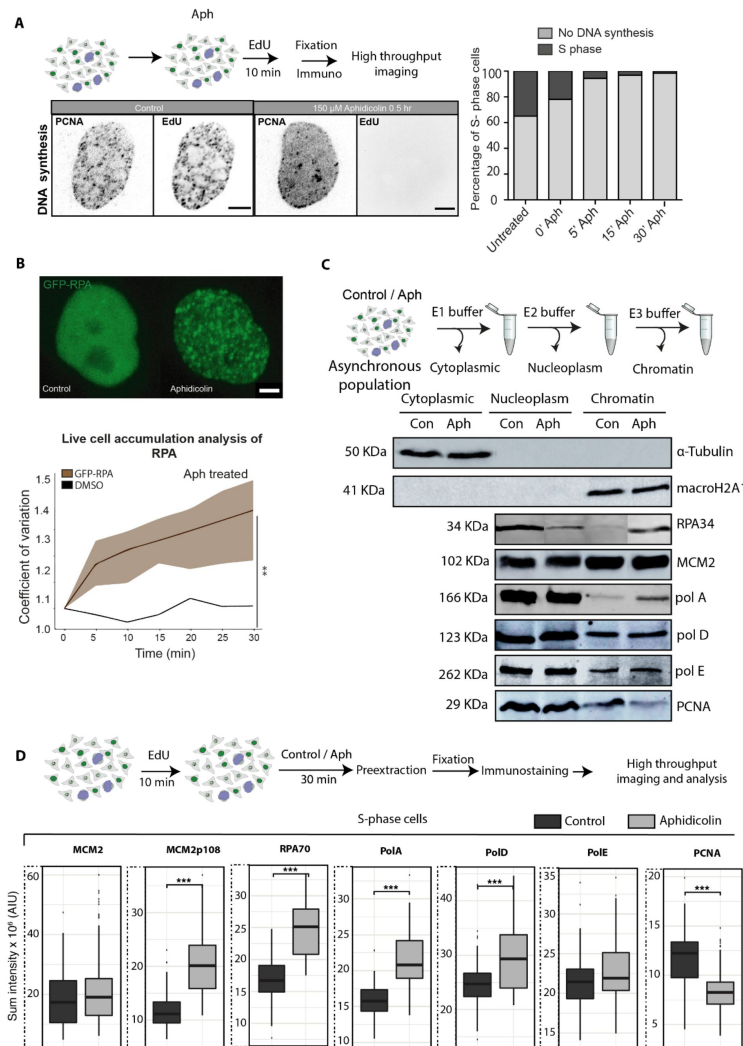


Figure 5

Dissecting the kinetics of replisome components after inhibition of DNA synthesis.

(A) The use of thymidine nucleotide analogs like 5-ethynyl-2'-deoxyuridine (EdU), which is incorporated into replicating DNA, allows us to estimate the time needed for complete inhibition of DNA synthesis. The representative images show no incorporation of EdU in S-phase cells upon aphidicolin (Aph) treatment for 30'. The plots below the images depict the % of cells with no DNA synthesis as scored by the EdU signal and the corresponding % of cells still replicating DNA (Supplementary Figures S4, S5). (B) The line plot shows the live cell accumulation analysis showing the normalized average RPA70 accumulation at replication sites (coefficient of variation $\pm$ standard deviation in transparent color) of HeLa cells stably expressing GFP-RPA34 (Supplementary Figures S6, S7, S8). (C) Western blots of cytoplasm, nucleoplasm and chromatin fractions of asynchronous population of HeLa cells probed for different replication factors. The western blots shown are cropped from the same replicates for easier visualization without contrast adjustment and the full blots are shown and highlighted in Supplementary Figure S9. (D) HeLa cells were pulsed with EdU for 10 min to identify S-phase cells and pre-extracted to detect chromatin bound proteins and different replication factors were detected using immunofluorescence. High-throughput imaging and image analysis were performed (Supplementary Figures S10, S11). Box plots depict the accumulation of the replisome factors indicated at DNA replication sites. Same Y-axis scale plots are shown in Supplementary Figure S11B. The boxplot lower and upper hinges correspond to the first and third quartiles (the 25th and 75th percentiles), the upper whisker extends from the hinge to the largest value no further than 1.5 $\times$ IQR from the hinge (where IQR is the interquartile range, or distance between the first and third quartiles). The lower whisker extends from the hinge to the smallest value at most 1.5 $\times$ IQR of the hinge. The horizontal line represents the median value. The outliers plotted individually as separate dots outside of the whiskers. *** p < 0.001 by Wilcoxon rank-sum test, for aphidicolin-treated versus control sample. Scale bar: 5 μ m.

GFP-RPA34 (**Supplementary Figure S6** [↗](#)). We performed time-lapse microscopy on HeLa GFP-RPA34 cells every 5 minutes for 60 min for both aphidicolin treated and control DMSO treated cells (**Supplementary Figure S7A**). We observed that RPA accumulated over time on DNA at replication sites in aphidicolin-treated cells but not in the control cells (**Supplementary Figures S7** [↗](#)). RPA accumulation indicated that the DNA helicase complexes continued unwinding the DNA, which allowed for increasing amounts of RPA to bind and, at the same time, the DNA polymerases were not active displacing the RPA while synthesizing the second (complementary) DNA strand (Görisch et al., 2008 [↗](#)). Therefore, we studied the kinetics of accumulation of RPA on chromatin upon DNA synthesis inhibition by quantifying the accumulation of GFP-RPA34 in live cells upon treatment with aphidicolin normalized to DMSO treated cells using the coefficient of variation (Cv), which indicates the amount of RPA protein accumulated over time (Methods, **Figure 5B** [↗](#), **Supplementary Figure S8** [↗](#)). We observed clear accumulation of RPA over time relative to control, showing that the single-strand DNA binding protein accumulates on chromatin. Hence, this indicates that upon stress the DNA helicase remained active unwinding the DNA.

Next, we analyzed the distribution of the helicase subunit MCM2 and its phosphorylated (p108) form along with DNA polymerases α , ϵ , and δ (**Supplementary Table 4** [↗](#)) at the chromatin. It has been previously described that the phosphorylated form of MCM2 is the active form for DNA unwinding (Forsburg, 2004 [↗](#); Montagnoli et al., 2006 [↗](#)). We predicted from the RPA accumulation that the helicase subunit was present at the replication sites and actively spooling the DNA through after the DNA synthesis inhibition. We first performed western blot analysis of different replication factors from asynchronous populations of HeLa cells after isolating the cytoplasm, nucleoplasm, and chromatin fractions (Methods). We tested the fractionation protocol by blotting the membranes with antibodies to α -tubulin for the cytoplasmic fraction and macro H2A1 histone for the chromatin fraction (**Figure 5C** [↗](#)). The same fractions were then incubated with antibodies for different replication factors. We observed significant dissociation of PCNA from chromatin and accumulation of RPA on chromatin upon aphidicolin treatment (**Figure 5C** [↗](#)) consistent with our fixed cell and live cell microscopy analysis. We found no significant changes in MCM2 helicase subunit levels on chromatin and higher levels of phosphorylation of MCM2 upon treatment with aphidicolin (**Figure 5C** [↗](#)). Lastly, we incubated the blots with antibodies recognizing the catalytic subunits of the DNA polymerases α , δ and ϵ complexes (Methods, **Supplementary Table 4** [↗](#)). The DNA polymerases showed a different behavior as compared to the DNA polymerase clamp protein, with DNA polymerase α being enriched on chromatin upon stress, with only minor to no changes being observed for the processive DNA polymerases δ and ϵ (**Figure 5C** [↗](#), 5D). It is of note that both these processive DNA polymerases bind the polymerase clamp PCNA whereas the far less processive DNA polymerase α does not. The full length blots are shown in the **Supplementary Figure S9** [↗](#).

We then performed an orthogonal analysis using high-throughput microscopy and image analysis. We labeled cells with EdU for 10 minutes to mark the S-phase cells and treated cells with DMSO/aphidicolin and subsequently performed pre-extraction to remove the unbound fraction of proteins and only detect the chromatin bound proteins. In this manner, we separately quantified accumulation only in S-phase cells and not in populations of cells including all cell cycle stages as in the previous western blot analysis (**Figure 5C** [↗](#)). The pre-extracted cells were fixed and immunostained for different replisome components (**Figure 5D** [↗](#)). The cells were imaged using a spinning disk confocal microscopy system (**Supplementary Table 5** [↗](#)) and image analysis was performed using the KNIME software with custom pipeline to quantify the accumulation/loss of replication factors on chromatin in S-phase cells (Methods, **Supplementary Table 9** [↗](#), **Supplementary Figures S10** [↗](#), **S11** [↗](#)). Using the EdU signal, the S-phase cells were selected for the quantitation of chromatin bound replisome components (**Figure 5D** [↗](#)). We found that PCNA dissociated from chromatin and RPA accumulated on chromatin upon stress in accordance with our previous analysis (**Figure 5C** [↗](#)). We found no changes in MCM2 helicase subunit but an increase in active MCM2p108 levels upon stress. This is consistent with no new loading of DNA helicases but de novo activation of already loaded helicase complexes (Ge et al., 2007 [↗](#); Ibarra et

al., 2008 [↗](#)). Finally, we observed significant accumulation of DNA polymerases α and δ on chromatin after aphidicolin treatment. PCNA does not associate with DNA polymerase α , which has a low processivity, but it associates with DNA polymerases ϵ and δ , which constitute the processive synthetic machinery responsible for most of the duplication of the genome. Hence, it was surprising that these two polymerases remain associated and even load *de novo* at non-synthesizing replication sites. The increase in DNA polymerase α could lead to the recruitment of alternative polymerase clamp 9-1-1 as reported before (Michael et al., 2000 [↗](#); Van et al., 2010 [↗](#); Yan and Michael, 2009a [↗](#), 2009b [↗](#)). Several scenarios explaining the different levels of DNA polymerases α and δ upon stress are possible (Supplementary Figure S12 [↗](#)): i) multiple polymerase complexes may load within the same Okazaki fragment, which is less likely in view of what is known on DNA replication (Supplementary Figure S12A); ii) multiple Okazaki fragments each with DNA polymerase α and δ within the same replication fork may form on the extended single stranded DNA unwound by the helicase complex (Supplementary Figure S12B); iii) additional replication origins may fire in the proximity of the stalled replication fork, which would explain the increase in both active phosphorylated helicase and DNA polymerases α and δ (Supplementary Figure S12C). Having established the conditions in which DNA synthesis but not DNA unwinding is blocked and concomitantly polymerases are accumulated, we then addressed the consequences for chromatin motion.

Accumulation of replisome components but not processive DNA synthesis *per se* restricts chromatin motion

To elucidate the roles of the processive DNA synthesis and loading of synthetic machinery in chromatin motion decrease in S-phase, we imaged single cells for PCNA, RPA34, and Cy3-dUTP pre and post aphidicolin treatment (Figure 6A [↗](#), see also movies 4,5). The PCNA and RPA patterns did not change in G1/G2, whereas in S-phase the PCNA was dissociated from chromatin and RPA was accumulated at the same previously replicating sites (Figure 6A [↗](#), see also movies 6,7). Chromatin motion analysis was performed on DNA labeled with Cy3-dUTP for both G1/G2 cells and S-phase cells pre and post-treatment with aphidicolin. We observed that chromatin motion was unaffected in G1/G2, which fit with our prediction, as in G1/G2 there is no active DNA synthesis besides possible DNA repair processes on a limited genomic scale (Figure 6B [↗](#)). As hydroxyurea, another DNA synthesis inhibitor, significantly affected chromatin mobility outside of S-phase, we did not further pursue it. Surprisingly, aphidicolin treatment and inhibition of DNA synthesis led to additional decrease in chromatin motion (Figure 6B [↗](#)) and the chromatin became even more constrained than at the proximity of the active replication sites in S-phase cells (see Figure 4C [↗](#)). As quantified above, after aphidicolin treatment, the helicases were still loaded and actively spooling DNA through, whereas the DNA polymerases α and δ albeit not synthesizing DNA accumulated on chromatin at the sites of helicase/RPA accumulation (Figure 5 [↗](#)).

In summary, we propose that the accumulation of the helicase and polymerase complexes on chromatin together with the continuous loading of the single stranded DNA binding protein (RPA) covering the ssDNA strands, stiffens the DNA polymer and restricts its diffusional motion. This study provides new insights on the kinetics of DNA replication proteins loading upon DNA replication stress and elucidates the transient and localized immobilization of chromatin during DNA replication.

Materials and Methods

Cells

All cells used were tested and negative for mycoplasma. Human cervical cancer cell line HeLa Kyoto (Erffle et al., 2007 [↗](#)) and human normal diploid fibroblasts from lung tissue IMR90 (Nichols et al. 1977 [↗](#)) were used in the study. Previously published HeLa Kyoto cells expressing GFP-PCNA

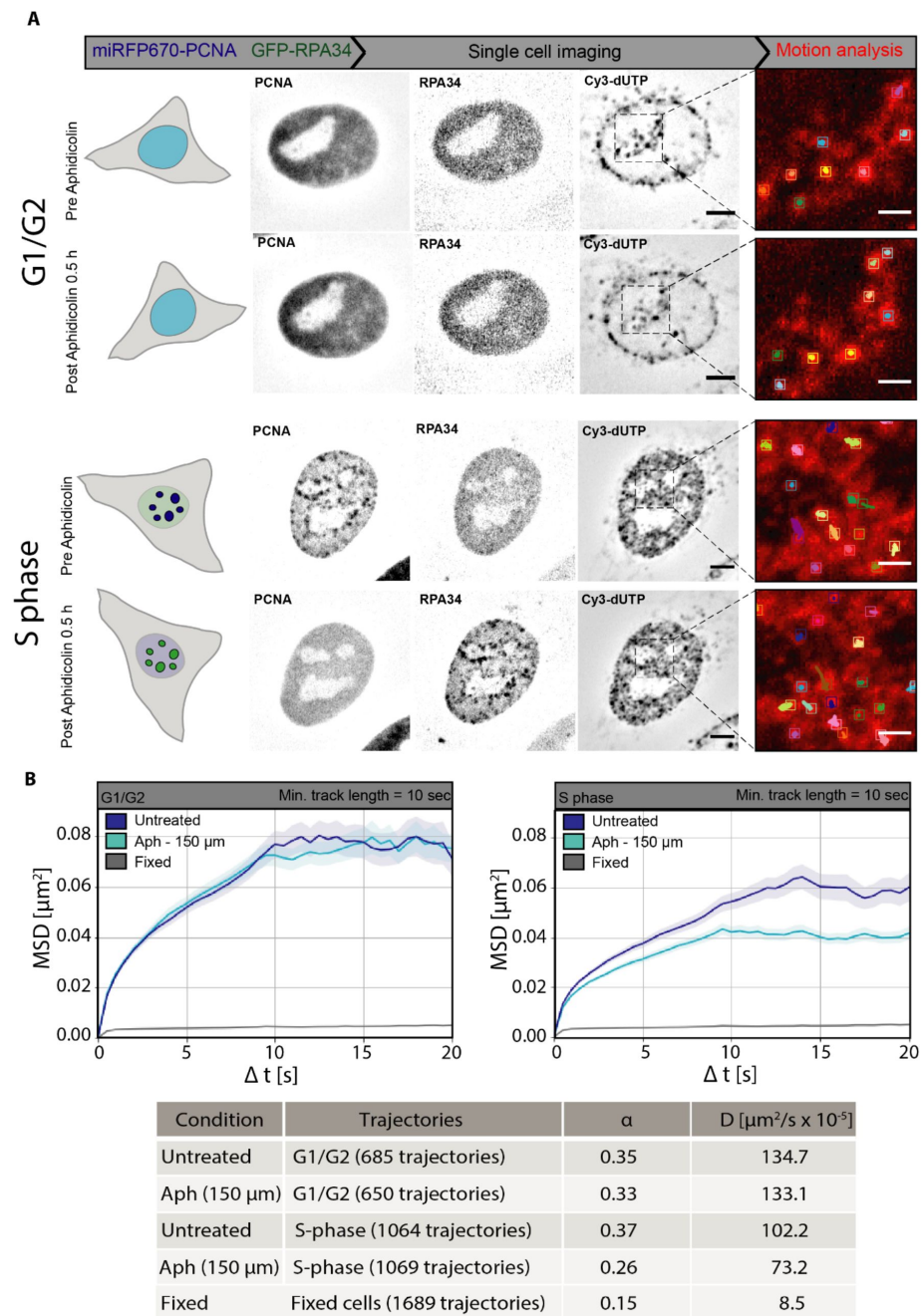


Figure 6

Inhibition of DNA synthesis by aphidicolin further restricts chromatin mobility in S phase but not in G1/G2 cells.

(A) Representative images of HeLa GFP-RPA34 cells transfected with a construct coding for miRFP670-PCNA and Cy3-dUTP nucleotides for both G1/G2 and S-phase cells pre and post aphidicolin (Aph) treatment. The chromatin foci were imaged using the spinning disk microscope. The image sequences were used to perform motion analysis. The cropped region (black dashed lines) show the motion analysis of chromatin tracks before and after treatment of the same cells (see also movies 6,7). (B) Mean square displacement curves over time were plotted for all chromatin tracks for untreated and aphidicolin-treated (150 μM) cells in G1/G2 and S-phase. The error bars are represented in transparent color around the curve. The table below provides the detailed information on number of trajectories per individual sample along with average diffusion rates ($\mu\text{m}^2/\text{s} \times 10^{-5}$) and anomalous α coefficient showing subdiffusion. The MSD's were plotted with error bars (standard deviation) represented in transparent color around the curve. Scale bar: 5 μm . Insets scale bar: 1 μm .

(Chagin et al., 2016 [↗](#)) fusion protein, were used to monitor cell cycle progression. HeLa Kyoto GFP-RPA34 were generated using the Flp-In recombination system based on the Flp site-specific recombinase (Cat.No.: K6010-01, Invitrogen, Waltham, Massachusetts, USA). The HeLa Kyoto FRTLacZ cells containing a genomically integrated FRT site described earlier (Chagin et al., 2016 [↗](#)) were cotransfected with pFRT-B-GRPA34 (**Supplementary Table 2** [↗](#)) (encoding GFP-RPA34) and pOG44 Flp-recombinase using Neon transfection (Cat.No.: MPK5000, Invitrogen, Waltham, Massachusetts, USA). Four hours after transfection the cell culture medium was exchanged and cells were grown for 48 h and selected with 2.5 mg ml⁻¹ blasticidin (Cat.No.: R210-01, Invitrogen, Waltham, Massachusetts, USA). A stable monoclonal line was isolated using blasticidin selection. All cells were maintained in Dulbecco's modified Eagle medium (DMEM) high glucose (Cat.No.: D6429, Sigma-Aldrich Chemie GmbH, Steinheim, Germany) supplemented with 10% fetal calf serum, 1x glutamine (Cat.No.: G7513, Sigma-Aldrich, St Louis, MO, USA) and 1 µM gentamicin (Cat.No.: G1397, Sigma-Aldrich, St Louis, MO, USA) in a humidified atmosphere with 5% CO₂ at 37 °C. Additional experiments confirmed that the transgenic gene product co-localized with the endogenous protein (not shown) and was present at sites of active replication (**Supplementary Figure S6** [↗](#)). The culture medium was changed every day and cells were split every two days. Cell line characteristics are summarized in **Supplementary Table 1** [↗](#).

To block DNA replication, cells were treated with aphidicolin (Aph) (Cat.No.: A0781-1MG, Sigma-Aldrich, St Louis, MO, USA) at final concentration of 150 µM (**Supplementary Table 3** [↗](#)). Cells were subsequently examined for 30 minutes (aphidicolin) following drug exposure. To confirm that DNA synthesis was inhibited, cells were labeled with 10 µM nucleoside analog 5-ethynyl-2'-deoxyuridine (Cat.No.: 7845.1, ClickIt-EdU cell proliferation assay, Carl Roth, Karlsruhe, Germany) (**Supplementary Table 3** [↗](#)) in media for 10 minutes to evaluate the extent of replication in control and treated cells (**Supplementary Figure S4** [↗](#)).

For synchronization of HeLa cells, the cells were seeded on tissue culture dishes at high confluency. Once the cells were confluent, the cells were placed on a shaker for 5 minutes. The detached mitotic cells were collected from the supernatant, and seeded on coverslips. Once the cells were in G1, they were fixed and stained with DAPI for quantification (**Supplementary Figure S2B**).

Live cell imaging and replication labeling

For live-cell microscopy, cells were transfected using a Neon transfection system (Cat.No.: MPK5000, Invitrogen, Waltham, Massachusetts, USA). Briefly, the asynchronous population of cells were washed with 1x PBS/EDTA, trypsinized, and collected in a 15 ml tube. The cells were pelleted at 300 x g for 5 minutes. The media was removed and cells were resuspended in 100 µl resuspension buffer R and transferred to a 1.5 ml microcentrifuge tube. Either 15 µg of plasmid DNA or/and 0.5 µl (25 nM) Cy3-dUTP (Cat.No.: ENZ-42501, Enzo Life Sciences, Lörrach, Germany) was added to the cell mixture (**Supplementary Tables 2** [↗](#), **3** [↗](#)). The NeonTM tip was immersed into the cell mixture and the mixture pipetted taking care to avoid bubbles. The tip was immersed in electrolytic buffer E2 and cells were electroporated (HeLa [voltage - 1005 V, width - 35, pulses - 2], IMR90 [1100 V, width - 30, pulses - 1]). The electroporated mixture was transferred to Ibidi µ-dish chambers (Cat.No.: 80826, Ibidi, Gräfelfing, Germany). Additionally, IMR90 cells were transfected with miRFP670-PCNA plasmid (**Supplementary Table 2** [↗](#)) to mark the DNA replication sites. After transfection, cells were allowed to attach overnight and were imaged the next day. All imaging was performed at 37 °C with a humidified atmosphere of 5% CO₂ using an Olympus environmental chamber (spinning disk microscope, **Supplementary Table 5** [↗](#)).

Immunofluorescence

For immunofluorescence, cells were fixed with 3.7 % formaldehyde/1x phosphate-buffered saline (PBS) (Cat.No.: F8775, Sigma-Aldrich Chemie GmbH, Steinheim, Germany) for 15 minutes and permeabilized with 0.7 % Triton-X100 in 1x PBS for 20 minutes. All washing steps were performed

with PBS-T (1x PBS/0.075 % Tween- 20). For detection of PCNA, cells were further incubated for 5 minutes in ice-cold methanol for antigen retrieval. Blocking (1% bovine serum albumin in 1x PBS) was performed for 30 minutes at room temperature. EdU was detected using the Click-IT assay as described by the manufacturer (1:1000 6-FAM azide or 1:2000 5/6-Sulforhodamine azide; Cat.No.: 7806 and 7776 respectively, Carl Roth, Karlsruhe, Germany). Primary and secondary antibodies were diluted in the blocking buffer and incubated for 1 hour at room temperature with subsequent 3×10 minutes of PBS-T washing. DNA was counterstained with DAPI (4',6-diamidino-2-phenylindole, 10 µg/ml, Cat.No.: D27802, Sigma-Aldrich Chemie GmbH, Steinheim, Germany) for 10 minutes, and samples were mounted in Vectashield (Cat.No.: VEC-H-1000, Vector Laboratories, Inc., Burlingame, CA, USA). Antibody characteristics are summarized in **Supplementary Table 4** [↗](#).

Western blot and chromatin fractionation

Cells for western blot were washed with 5 ml ice-cold 1x PBS once and 2 ml of ice-cold 1x PBS was added and cells were scraped using a cell scraper. Cells were then centrifuged in a 15 ml tube at 500 x g for 5 minutes. Cells were lysed for total cell lysates for 1 hr at 4 °C using the IP lysis buffer with (150 mM NaCl (Cat.No.: 0601.2, Carl Roth, Karlsruhe, Germany), 200 mM TrisCl pH 8 (Cat.No.: A1086.500, Diagonal, Münster, Germany), 5 mM EDTA (Cat.No.: 8040.2, Carl Roth, Karlsruhe, Germany), 0.5 % NP-40 (Cat.No.: 74385, Sigma-Aldrich Chemie GmbH, Steinheim, Germany)) and protease and phosphatase inhibitors (PMSF (Cat.No.: 6367.1, Carl Roth, Karlsruhe, Germany), PepA (Cat.No.: 2936.2, Carl Roth, Karlsruhe, Germany), NaF (Cat.No.: 67414-1ML-F, Sigma-Aldrich Chemie GmbH, Steinheim, Germany), Na₃VO₄ (Cat.No.: S6508-10G, Sigma-Aldrich Chemie GmbH, Steinheim, Germany)). Protein fractionation of control and treated samples was performed as described in (Gillot, 2018 [↗](#)). Briefly, equal number of cells were washed with buffer E1 (cytoplasmic fraction) and centrifuged at 1200 x g for 2 min and collected into a new tube. The step was repeated 2 times to remove excess cytoplasmic fraction. The pellet was then washed with buffer E2 (nucleoplasm fraction) and collected into a new tube. The chromatin fraction was isolated with buffer E3 and 1:1000 benzonase for 20 minutes at 25 °C.

All lysates was then centrifuged at 13,000 rpm for 20 minutes at 4 °C. The supernatant was collected into a new 1.5 ml tube and protein concentration was measured using the bovine serum albumin protein standard assay (Cat.No.: 23208, Thermo Fisher Scientific, Waltham, Massachusetts, USA) according to the manufacturer's protocol. 10 % SDS-PAGE gel was prepared and 50 µg of protein lysate was loaded along with the protein standard ladder (Cat.No.: P7719S, New England Biolabs, Ipswich, Massachusetts, United States), and electrophoresis was performed for 1.5 hr in ice-cold 1x Laemmli electrophoresis running buffer. Then, the protein was transferred to the 0.2 µm nitrocellulose membrane using a semi-dry transfer system (#1703940, Trans-Blot® SD Semi-Dry Transfer Cell, Bio-Rad, Hercules, CA, USA) for 55 min at 25 V using 1x transfer buffer (Pierce Western Blot Transfer Buffer 10x, Thermo Fisher Scientific, Waltham, Massachusetts, USA). After the transfer, the blotting membrane was incubated in a blocking buffer (5 % low-fat milk in 1x PBS) for 30 minutes. The primary antibodies (**Supplementary Table 4** [↗](#)) were diluted in blocking buffer to 5 % milk and incubated at 4 °C overnight. The next day the membrane was washed 3 times with 1x PBS-T (0.075 %) 10 minutes each. The membrane was then incubated with secondary antibodies (**Supplementary Table 4** [↗](#)) for 1 hr at room temperature. The membrane was washed again with 1x PBS-T (0.075 %) 3 times 10 minutes each and incubated with 1:1 ECL chemiluminescence solution (Clarity Western ECL, #170-5061, Bio-Rad Laboratories, Hercules, CA, USA). Signal was detected using an Amersham AI600 imager (**Supplementary Table 5** [↗](#)).

Microscopy

Live cell imaging for chromatin mobility measurements were performed using the PerkinElmer UltraVIEW VoX system with a 60x/1.45 numerical aperture plan-apochromatic oil immersion objective. Cy3 and GFP were excited sequentially using 543 nm and 488 nm solid-state diode laser lines to minimize crosstalk. The standard protocol for examining chromatin mobility in Cy3-dUTP

labeled nuclei was as follows: first, a reference image comprising the mRFP670/GFP-PCNA, Cy3-dUTP, and the phase-contrast signal was collected from a single focal plane corresponding to the middle of the nucleus. This image demarcated the nuclear boundary, provided cell cycle information, and, in the case of S-phase cells, allowed us to correlate the positions of Cy3-dUTP foci with sites of DNA replication. Second, while maintaining the same focal plane, a time series (30-60 seconds) at a frame rate of 500 ms was captured. To maximize the temporal resolution, the time series consisted solely of the Cy3-dUTP channel and a PCNA reference frame at the beginning to obtain information on the cell cycle stage.

Multiple point time-lapse microscopy was performed using the multi-time option available in the spinning disk Volocity 6.3 software to image the chromatin (Cy3-dUTP) of the same cells pre and post treatment of aphidicolin. To minimize photo-toxicity over the course of the experiment, transmitted light contrast imaging was used to focus the cells. Live cell imaging was performed by following cells through the cell cycle and G1 and G2 stages were classified based on the previous cell cycle stage.

For the inhibition experiments (aphidicolin) different cells/points were chosen using the multipoint function of the Perkin Elmer spinning disk, and image sequences before the treatment were acquired. The reference image consisted of GFP-RPA34, Cy3-dUTP, and mRFP670-PCNA using 488 nm, 561 nm, and 640 nm solid-state diode lasers, respectively. After acquiring the reference images the media containing the small molecule inhibitor was added to cells on the microscope for the required time and after treatment image sequences were acquired for analysis of chromatin motion.

High-throughput imaging was performed using the 40x/0.95 numerical aperture air objective of the PerkinElmer Operetta system. We used different filters (excitation/emission: 360/400, 460/490, 560/580) to image DAPI, EdU, and different replication proteins. (Figure 5 [↗](#), Supplementary Figure S5 [↗](#), Supplementary Table 5 [↗](#)).

Quantification of DNA synthesis inhibition

The high-throughput images were used to quantify the percentage of cells with inhibition of DNA synthesis upon aphidicolin treatment. A minimum of 100 fields with around 2000-5000 cells were acquired in all channels. The images were then analyzed using the PerkinElmer Harmony software. The steps in brief (Supplementary Figure S4 [↗](#), S5 [↗](#)) include segmentation of nuclei using cell types specific parameters like the diameter, splitting coefficient, and intensity threshold. The segmentation was then validated by visually checking it in randomly selected regions. Once the nuclei were segmented, cells touching the border were omitted. The intensity values with mean, median, standard deviation, and the sum of the intensities were obtained for individual cells. The datasheets were then imported to R and plots were generated. EdU signal was used to identify the population of cells actively replicating upon Aph treatment (Figure 5A [↗](#)). The background intensity for EdU staining was determined using a negative control which was not treated with EdU but stained. The cells showing a mean intensity greater than the background intensity were separated into an EdU positive population and plotted.

DNA quantification of labeled chromatin

DNA quantification of the labeled foci was done by automated image analysis. Image sequences with labeled chromatin were acquired on a Ultra-View VoX spinning disk microscope, using a 60x objective (Figure 2 [↗](#), Supplementary Figure S1 [↗](#)). For segmentation of replication foci, we used the protocol originally described in (Chagin et al., 2016 [↗](#), 2015 [↗](#)). The channels comprising DAPI replication foci signals were imported into the software Perkin Elmer Volocity 6.3 and converted into volumes. The pixel dimensions of the images were set to the specifications for the spinning disk (x/y: 0.066 μ m and z: 0.3 μ m). The following processing steps were applied: Find objects (“nucleus”) using the DAPI channel, method “Intensity” (set manually to the optimal value), use fill

holes in object/dilate/erode until the object optimally fits the nucleus, exclude objects by size $< 500 \mu\text{m}^3$. Find objects using the label channel, method “Intensity” (lower limit: 1, upper limit: 65535), separate touching objects, exclude “foci” not touching “nucleus”. Using the detected foci, the DNA content of foci was determined via the sum of intensities in the DAPI channel and the genome size of the cell type (**Figure 2**, **Supplementary Figure S1**).

Automated tracking of chromatin structures in time-lapse videos

The motility of fluorescently labeled chromatin structures in live-cell fluorescence microscopy images was quantified within manually segmented single nuclei. The background image intensity was adjusted for each image sequence to the computed mean intensity value over all time points within a manually selected region of interest (ROI) of the background. Automatic tracking of multiple fluorescently labeled chromatin structures was performed using a probabilistic particle tracking approach, which is based on Bayesian filtering and multi-sensor data fusion (Ritter et al., 2021). This approach combines Kalman filtering with particle filtering and integrates multiple measurements by separate sensor models and sequential multi-sensor data fusion. Detection-based and prediction-based measurements are obtained by elliptical sampling (Godinez and Rohr, 2015), and the separate sensor models allow taking into account different uncertainties. In addition, motion information based on displacements from past time points is exploited and integrated in the cost function for correspondence finding. Chromatin structures are detected by the spot-enhancing filter (SEF) (Sage et al., 2005) which consists of a Laplacian-of-Gaussian (LoG) filter followed by thresholding the filtered image and determination of local maxima. The threshold is automatically determined by the mean of the absolute values of the filtered image plus a factor times the standard deviation. We used the same threshold factor for all images of an image sequence (**Supplementary Figure S2**).

Chromatin motility analysis

Based on the computed trajectories, the motility of chromatin structures was analyzed and the motion type was determined for different cell cycle stages along with active replication sites, and inhibition of DNA synthesis with aphidicolin. We performed a mean square displacement (MSD) analysis (Saxton, 1997) and computed the MSD as a function of the time interval Δt for each trajectory (**Supplementary Figure S2**). The MSD curves for all trajectories with a minimum time duration of 10 s (corresponding to 20 time steps) under one condition were averaged. We considered only the trajectories with a time duration larger than the minimum time duration which improved the accuracy of the motility analysis. We fitted the anomalous diffusion model to the calculated MSD values to obtain the anomalous diffusion coefficient α . The motion was classified into confined diffusion, obstructed diffusion, and normal diffusion (Bacher et al., 2004). To determine the diffusion coefficient D [$\mu\text{m}^2\text{s}^{-1}$], the diffusion model was fitted to the MSD values. In case of IMR90 cells, affine image registration was performed using the method in (Celikay et al., 2022) to address the stronger cell movement compared to HeLa cells.

Center to center distance / Proximity analysis

Automatic proximity analysis of chromatin and PCNA was performed using the computed trajectories of chromatin structures and detected sites of active DNA synthesis represented by fluorescently labeled PCNA. Only trajectories of chromatin structures present at the first time point of an image sequence and with a minimum time duration of 10 s (corresponding to 20 time steps) were considered. PCNA foci were automatically detected in the fluorescence microscopy images by the SEF filter (**Supplementary Figure S2**). For each PCNA image, a single nucleus was manually segmented and the background intensity was adjusted to the computed mean intensity value within a manually selected ROI of the background. Proximity was determined for the first time point of the trajectory of a chromatin structure and detected PCNA foci using a graph-based k -d-tree approach (Bentley, 1975). Due to the k -d-tree structure, this approach allows efficient computation of the nearest neighbor query based on the Euclidean distance between foci in the

chromatin and PCNA channel. If a chromatin structure at the first time point of the image sequence has a nearest PCNA neighbor within a maximum distance, the trajectory of a chromatin structure is considered within center to center distance (CCD). Otherwise, the trajectory is considered outside the center to center distance (CCD) (**Supplementary Figure S3** [↗](#)).

Accumulation analysis

To analyze the focal RPA accumulation upon DMSO/aphidicolin treatment, cell nuclei were segmented using the Volocity software (Version 6.3, Perkin Elmer). The GFP-RPA34 signal was segmented before and after treatment of the same cell in the live experiments and plotted over time after DMSO and drug treatment. The GFP-RPA intensities were measured and the coefficient of variation $cV = \sigma/\mu$, with σ = standard deviation and μ = mean) was calculated for all time points (**Supplementary Figures S7** [↗](#), **S8** [↗](#)). All values were normalized to the DMSO treatment $cV = cV(tpx)/cV(tp0)$ with tpx : any given time point imaged, $tp0$: pretreatment time point).and plotted using RStudio (**Supplementary Table 9** [↗](#)).

High throughput image analysis of replisome components

The images from the Nikon crest Ti2 system were analyzed with the custom made image analysis pipeline in KNIME Analytics Platform. The image analysis pipeline was constructed as follows (**Supplementary Figures S10** [↗](#), **S11** [↗](#)). Briefly, the channels were separated. The DAPI channel was used for the nuclei segmentation. Nuclei were segmented based on manually chosen intensity threshold, the Watershed Transform was applied next to separate the close-positioned nuclei. The segmented nuclei were converted into a mask with each nucleus DAPI intensity and texture features recorded. The nuclei population was further thresholded by nucleus area and circularity to eliminate segmentation artifacts. The EdU and replication protein channels were subjected to foci segmentation based on a wavelet transform algorithm. The algorithm parameters were selected individually for each type of the replication protein and maintained the same between the control and treated samples. The nuclear mask and EdU foci/replication protein foci masks were overlaid to filter only the foci inside the nuclear areas. The EdU foci/replication protein foci intensity parameters (total focus intensity, mean focus intensity), area and foci number per nucleus were exported as XLSX files for further analysis. The data was analyzed in R Studio (<https://posit.co/download/rstudio-desktop/> [↗](#)). First, the S phase cell population was identified by the number of EdU foci per nucleus. The EdU foci number threshold was set as 50 for the cells in control samples, and 55 for aphidicolin-treated samples among all datasets. The nuclei in S phase were next analyzed for their replication protein accumulation. The total levels of the replication proteins was plotted as boxplots, *** $p < 0.001$ by Wilcoxon rank-sum test, for aphidicolin-treated vs. control sample.

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Data Availability Statement

All data are available from the OMERO open microscopy environment public repository <http://cc-omero.bio.tutudarmstadt.de/webclient/userdata/?experimenter=-1> (<https://doi.org/10.48328/tudatalib-873>). All renewable biological materials will be made available upon request from the corresponding author M. Cristina Cardoso (cardoso@bio.tu-darmstadt.de).

Author contributions

M.K.P., C.R., V.O.C. and M.C.C. designed the study and wrote the manuscript. K.R., and H.L., discussed the project and contributed to the manuscript writing. M.K.P., C.R., V.O.C., J.M., K.C., J.H.S., D.L., K.K., P.P. and A.K.S contributed to experiments, data analysis and figure preparation. All authors reviewed the manuscript.

Declaration of interests

The authors declare no competing interests.

Supplementary tables and figures

Movie captions

Movie 1: Time lapse microscopy of HeLa K cells in G1 phase expressing fluorescent PCNA (green) and labeled chromatin (red). Scale bar: 5 μ m.

Movie 2: Time lapse microscopy of HeLa K cells in G2 phase expressing fluorescent PCNA (green) and labeled chromatin (red). Scale bar: 5 μ m.

Movie 3: Time lapse microscopy of HeLa K cells in S phase expressing fluorescent PCNA (green) and labeled chromatin (red). Scale bar: 5 μ m.

Movie 4: Time lapse microscopy of IMR90 cells in G1 phase expressing fluorescent PCNA (green) and labeled chromatin (red). Scale bar: 5 μ m.

Movie 5: Time lapse microscopy of IMR90 cells in S phase expressing fluorescent PCNA (green) and labeled chromatin (red). Scale bar: 5 μ m.

Movie 6: Time lapse microscopy of HeLa K cells pre and post aphidicolin treatment in G1/G2 phase expressing fluorescent PCNA and RPA and labeled chromatin (red). Scale bar: 5 μ m.

Movie 7: Time lapse microscopy of HeLa K cells pre and post aphidicolin treatment in S phase expressing fluorescent PCNA and RPA and labeled chromatin (red). Scale bar: 5 μ m.

Name	Species	Type	Genome size (Gbp)	Reference
HeLa K	Homo sapiens	Cervical adenocarcinoma	9.682±0.002	(Erflie et al., 2007)
HeLa K GFP-PCNA	Homo sapiens	Cervical adenocarcinoma	9.682±0.002	(Chagin et al., 2016)
HeLa K GFP-RPA34	Homo sapiens	Cervical adenocarcinoma	9.682±0.002	This study
HeLa K FRTLacZ	Homo sapiens	Cervical adenocarcinoma	9.682±0.002	(Chagin et al., 2016)
IMR90	Homo sapiens	Fibroblasts from lung tissue	6.37	(Nichols et al. 1977)

Supplementary table 1

Cell line characteristics

Name	pc number	Fluorophore	Protein of Interest	Promoter	References
pmiRFP670-PCNA	3385	miRFP670	Human PCNA	CMV	(Rausch et al., 2021)
pFRT-B-GRPA	1232	GFP	Human RPA34	EF1 α	This study
pFRT-B-GPCNA	1274	GFP	Human PCNA	EF1 α	Chagin et al., 2016)

* pc: plasmid collection.

Supplementary table 2

Plasmid characteristics

Name	Application	Detection	Cat #	Company
Cy3-dUTP	Replication labeling (Labeling of nascent DNA)	-	ENZ-42501	Enzo life sciences, Farmingdale, NY, USA
Aphidicolin	Replisome disruption by polymerase inhibition	-	A0781-1MG	Sigma-Aldrich, St Louis, MO, USA
5-ethynyl-2'- deoxyuridine (EdU)	Labeling of nascent DNA in pulse (chase) experiments	ClickIT chemistry	E10415	Thermo Fisher Scientific, Waltham, MA, USA

Supplementary table 3

Nucleotide and chemical characteristics

Reactivity	Host	Clonality	Dilution	Application	Cat / Clone [#]	Company / References
anti RPA34	Mouse	Monoclonal	1:2	IF, WB	9H8H4 [#]	Gift from Mark Kenny/J. Hurwitz (Kenny et al., 1990)
anti RPA70A	Mouse	Monoclonal	1:2	IF, WB	7G9E3 [#]	Gift from Mark Kenny/J. Hurwitz (Kenny et al., 1990)
Anti MCM2	Rabbit	Monoclonal	1:5000 IF, 1:10000 WB	IF, WB	ab 108935/ EPR4120	Abcam, Cambridge, United Kingdom
anti MCM2pS108	Rabbit	Monoclonal	1:1000	IF, WB	3267-1	Epitomics, Burlingame, CA, United States
anti pol Alpha	Mouse	Monoclonal	Undiluted	IF, WB	SJK-287-38 [#]	ATCC (Tanaka et al., 1982)
anti pol Delta	Mouse	Monoclonal	1:500	IF, WB	610972	BD biosciences, New Jersey, USA (Li et al., 2016)
anti pol Epsilon	Rabbit	Polyclonal	1:500	IF, WB	GTX132100	GeneTex, Irvine, California, United States
anti PCNA	Mouse	Monoclonal	1:100	IF*, WB	M0879 / PC10 [#]	Dako, Hamburg, Germany (Waseem and Lane, 1990)
anti Histone H3	Rat	Monoclonal	1:250	WB	61647/ 1C8B2 [#]	Active Motif, California, USA
anti MacroH2A1	Rabbit	Polyclonal	1:1000	WB	07-219	Active Motif, California, USA

Supplementary table 4

Primary and secondary antibody characteristics

anti GFP	Rat	Monoclonal	1:1000	WB	3H9#	Chromotek 3H9, Planegg-Martinsried, Germany
Anti tubulin alpha	Mouse	Monoclonal	1:5000	WB	clone DM1A#/ T9026	Sigma, Missouri, United States
anti-mouse IgG Cy3	Donkey	Polyclonal	1:800	IF (fluorescent secondary)	715-165-151	The Jackson Laboratory, Bar Harbor, ME, USA
anti-rabbit IgG Cy3	Donkey	Polyclonal	1:800	IF (fluorescent secondary)	711-165-152	The Jackson Laboratory, Bar Harbor, ME, USA
anti-mouse IgG Cy5	Donkey	Polyclonal	1:800	F (fluorescent secondary)	715-175-150	The Jackson Laboratory, Bar Harbor, ME, USA
anti-rabbit IgG Cy5	Donkey	Polyclonal	1:800	F (fluorescent secondary)	711-175-152	The Jackson Laboratory, Bar Harbor, ME, USA
anti-mouse IgG AF488	Goat	Polyclonal	1:800	IF (fluorescent secondary)	2120125	Invitrogen, Waltham, Massachusetts, USA
anti-rabbit IgG AF488	Donkey	Polyclonal	1:800	IF (fluorescent secondary)	A11034	Invitrogen, Waltham, Massachusetts, USA
anti rat IgG HRP	Goat	Polyclonal	1:5000	WB (HRP secondary)	112-035-068	The Jackson Laboratory, Bar Harbor, ME, USA
anti mouse IgG HRP	Sheep	Polyclonal	1:5000	WB (HRP secondary)	NA931	Amersham pharmacia, Amersham, United Kingdom

*Methanol treatment required #Clone number

Supplementary table 4 (continued)

Microscope/ Company	Lasers/lamps	Filters (ex. & em. [nm])*	Objectives/ lenses	Detection system	Incubation system	Application
Ultra-View VoX spinning disk microscope/ PerkinElmer Life Sciences, UK	solid state diode lasers (405 nm, 488 nm, 561 nm, 640 nm)	405/488/56 8/640** 405: 415–475 488: 505–549 561: 580–650 640: 664–754	oil immersion 60x Plan- Apochromat (NA 1.45)	cooled 14-bit Hamamatsu® C9100-50 EMCCD	closed live- cell microscopy chamber (ACU control, Olympus) for time-lapse microscopy	time-lapse microscopy & confocal z- stack imaging
Widefield microscope Axiovert 200 /Zeiss, Germany	HBO100 mercury lamp	488: 473-491 & 506-534 561: 550-580 & 590-650 640: 590-650 & 663-738	oil immersion 63x Plan- Apochromat (NA 1.4)	12-bit AxioCam mRM	-	Multi channel wide-field imaging
Leica SP5 II confocal microscope /Wetzlar, Germany	405 nm diode 488 nm Argon, 561 nm DPSS, 633 nm HeNe	AOBS beam splitter	HCX PL APO 63x / 1.4-0.6 oil lambda blue & HCX PL APO 100x (NA 1.44) oil Corr CS	2 HyD Hybrid Detectors	-	confocal z- stack imaging
Amersham Al600 imager	Chemiluminesce nce, UV transillumination	-	large aperture FUJINON™ f/0.85 43 mm	16-bit Peltier cooled Fujifilm Super CCD	-	Western blots and DNA agarose gels
Operetta high throughput imaging/ PerkinElmer Life Sciences, UK	Xenon fiber-optic light source, 300 W, 360 – 640 nm continuous spectrum LED light source for transmission mode	ex:360/400, 460/490, 560/580 em: 410/480, 500/550, 560/630	20x or 40x air (0.45 NA and 0.95 NA) long WD***	14 bit Jenoptik firecamj203 Sony Chip ICX285 cooled 20°C below environment	-	high throughput, high content imaging and image analysis

Supplementary table 5

Imaging systems characteristics

Nikon TiE2 inverted with crest spinning disk unit/ Nikon, Japan	SPECTRA X light engine 395/25 nm with 295 mW 440/20 nm with 256 mW 470/24 nm with 196 mW 510/25 nm with 62 mW 540/30 nm with 231 mW 550/15 nm with 260 mW 575/25 nm with 310 mW	LED- DA/FI/TR/Cy5- 4X-B Quadbandpasse x:390/18, 475/35, 535/50 em:460/60, 530/43, 580LP	40x air (0.95 NA) & 250 µm WD***	Cooled Nikon Qi2 camera and 16.25 megapixel sCMOS sensor. readout noise is: 2.2. electron	-	high throughput, high content imaging and image analysis
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* ex.: excitation & em.: emission, ** dichroic specification, *** WD: working distance.

Supplementary table 5 (continued)

Name	Cell stage	DAPI SUM	Correction factor (C)	Corrected genome size (GSxC) Gbp
Cell1_crop_DAPI_Cy3dUTP_HeLa	SE	6.55E+08	1.05	10.185
Cell2_crop_DAPI_Cy3dUTP_HeLa	SE	8.49E+08	1.05	10.185
Cell3_crop_DAPI_Cy3dUTP_HeLa	SE	9.68E+08	1.05	10.185
Cell4_crop_DAPI_Cy3dUTP_HeLa	SM	1.14E+09	1.25	12.125
Cell5_crop_DAPI_Cy3dUTP_HeLa	SL	1.15E+09	1.77	17.169
Cell6_crop_DAPI_Cy3dUTP_HeLa	SE	8.29E+08	1.05	10.185
Cell7_crop_DAPI_Cy3dUTP_HeLa	SE	9.49E+08	1.05	10.185
Cell8_crop_DAPI_Cy3dUTP_HeLa	SM	1.01E+09	1.25	12.125
Cell9_crop_DAPI_Cy3dUTP_HeLa	SM	1.17E+09	1.25	12.125
Cell10_crop_DAPI_Cy3dUTP_HeLa	SM	1.00E+09	1.25	12.125

Supplementary table 6

Data description for DNA quantification

Cell11_crop_DAPI_Cy3dUTP_HeLa	SL	1.34E+09	1.77	17.169
Cell12_crop_DAPI_Cy3dUTP_HeLa	SM	1.06E+09	1.25	12.125
Cell13_crop_DAPI_Cy3dUTP_HeLa	SL	1.49E+09	1.77	17.169
Cell14_crop_DAPI_Cy3dUTP_HeLa	SE	8.71E+08	1.05	10.185
Cell15_crop_DAPI_Cy3dUTP_HeLa	SL	1.18E+09	1.77	17.169
Cell16_crop_DAPI_Cy3dUTP_HeLa	SL	1.48E+09	1.77	17.169
Cell17_crop_DAPI_Cy3dUTP_HeLa	SL	1.09E+09	1.77	17.169
Cell18_crop_DAPI_Cy3dUTP_HeLa	SL	1.07E+09	1.77	17.169
Cell19_crop_DAPI_Cy3dUTP_HeLa	SL	1.17E+09	1.77	17.169
Cell20_crop_DAPI_Cy3dUTP_HeLa	SM	1.09E+09	1.25	12.125
Cell21_crop_DAPI_Cy3dUTP_HeLa	SL	1.02E+09	1.77	17.169
Cell22_crop_DAPI_Cy3dUTP_HeLa	SL	1.02E+09	1.77	17.169
Cell23_crop_DAPI_Cy3dUTP_HeLa	SE	9.33E+08	1.05	10.185
Cell24_crop_DAPI_Cy3dUTP_HeLa	SE	8.90E+08	1.05	10.185
Cell25_crop_DAPI_Cy3dUTP_HeLa	SE	7.67E+08	1.05	10.185
Cell26_crop_DAPI_Cy3dUTP_HeLa	SE	8.39E+08	1.05	10.185
Cell27_crop_DAPI_Cy3dUTP_HeLa	SM	1.04E+09	1.25	12.125
Cell28_crop_DAPI_Cy3dUTP_HeLa	SM	1.01E+09	1.25	12.125
Cell29_crop_DAPI_Cy3dUTP_HeLa	SM	1.05E+09	1.25	12.125

Supplementary table 6 (continued)

Cell30_crop_DAPI_Cy3dUTP_HeLa	SM	1.02E+09	1.25	12.125
Cell1_DAPI_Cy3dUTP_IMR90	-	1.08E+09	-	6.37
Cell2_DAPI_Cy3dUTP_IMR90	-	5.04E+08	-	6.37
Cell3_DAPI_Cy3dUTP_IMR90	-	1.09E+09	-	6.37
Cell4_DAPI_Cy3dUTP_IMR90	-	8.72E+08	-	6.37
Cell5_DAPI_Cy3dUTP_IMR90	-	1.08E+09	-	6.37
Cell6_DAPI_Cy3dUTP_IMR90	-	5.39E+08	-	6.37
Cell7_DAPI_Cy3dUTP_IMR90	-	6.33E+08	-	6.37
Cell8_DAPI_Cy3dUTP_IMR90	-	5.52E+08	-	6.37
Cell9_DAPI_Cy3dUTP_IMR90	-	6.20E+08	-	6.37
Cell10_DAPI_Cy3dUTP_IMR90	-	6.68E+08	-	6.37
Cell13_DAPI_Cy3dUTP_IMR90	-	7.43E+08	-	6.37
Cell14_DAPI_Cy3dUTP_IMR90	-	6.53E+08	-	6.37
Cell15_DAPI_Cy3dUTP_IMR90	-	7.25E+08	-	6.37
Cell16_DAPI_Cy3dUTP_IMR90	-	4.41E+08	-	6.37
Cell17_DAPI_Cy3dUTP_IMR90	-	1.29E+09	-	6.37
Cell18_DAPI_Cy3dUTP_IMR90	-	7.51E+08	-	6.37

Supplementary table 6 (continued)

Name	Cell stage	Time (s)	Channels*	Pixel size (nm)	frame rate (ms)
Cell1_HeLa_fixedcells.tif	SE	40	3	120	500
Cell2_HeLa_fixedcells.tif	SE	40	3	120	500
Cell3_HeLa_fixedcells.tif	SE	40	3	120	500
Cell4_HeLa_fixedcells.tif	SM	40	3	120	500
Cell5_HeLa_fixedcells.tif	SM	40	3	120	500
Cell6_HeLa_fixedcells.tif	SM	40	3	120	500
Cell7_HeLa_fixedcells.tif	SM	40	3	120	500
Cell8_HeLa_fixedcells.tif	SL	40	3	120	500
Cell9_HeLa_fixedcells.tif	SL	40	3	120	500
Cell10_HeLa_fixedcells.tif	SL	40	3	120	500
Cell11_HeLa_fixedcells.tif	SE	40	3	120	500
Cell12_HeLa_fixedcells.tif	SE	40	3	120	500
Cell13_HeLa_fixedcells.tif	SE	40	3	120	500
Cell14_HeLa_fixedcells.tif	SL	40	3	120	500
Cell15_HeLa_fixedcells.tif	SE	40	3	120	500
Cell16_HeLa_fixedcells.tif	SE	40	3	120	500
Cell17_HeLa_fixedcells.tif	SL	40	3	120	500

Supplementary table 7

Data description for 2D confocal fixed images

Cell18_HeLa_fixedcells.tif	G2	40	3	120	500
Cell19_HeLa_fixedcells.tif	G1	40	3	120	500
Cell20_HeLa_fixedcells.tif	SM	40	3	120	500
Cell21_HeLa_fixedcells.tif	SE	40	3	120	500
Cell22_HeLa_fixedcells.tif	SE	40	3	120	500
Cell23_HeLa_fixedcells.tif	SL	40	3	120	500
Cell24_HeLa_fixedcells.tif	SM	40	3	120	500
Cell25_HeLa_fixedcells.tif	SM	40	3	120	500
Cell26_HeLa_fixedcells.tif	SE	40	3	120	500
Cell27_HeLa_fixedcells.tif	SE	40	3	120	500
Cell28_HeLa_fixedcells.tif	SE	40	3	120	500
Cell29_HeLa_fixedcells.tif	SM	40	3	120	500
Cell30_HeLa_fixedcells.tif	SE	40	3	120	500
Cell31_HeLa_fixedcells.tif	SL	40	3	120	500
Cell32_HeLa_fixedcells.tif	SE	40	3	120	500
Cell33_HeLa_fixedcells.tif	SE	40	3	120	500
Cell34_HeLa_fixedcells.tif	SE	40	3	120	500
Cell35_HeLa_fixedcells.tif	SE	40	3	120	500
Cell1_IMR90_fixed.tif	-	40	3	120	500

Supplementary table 7 (continued)

Cell2_IMR90_fixed.tif	-	40	3	120	500
Cell3_IMR90_fixed.tif	-	40	3	120	500
Cell4_IMR90_fixed.tif	-	40	3	120	500
Cell5_IMR90_fixed.tif	-	40	3	120	500
Cell6_IMR90_fixed.tif	-	40	3	120	500
Cell7_IMR90_fixed.tif	-	40	3	120	500
Cell8_IMR90_fixed.tif	-	40	3	120	500
Cell9_IMR90_fixed.tif	-	40	3	120	500
Cell10_IMR90_fixed.tif	-	40	3	120	500
Cell11_IMR90_fixed.tif	-	40	3	120	500
Cell12_IMR90_fixed.tif	-	40	3	120	500
Cell13_IMR90_fixed.tif	-	40	3	120	500
Cell14_IMR90_fixed.tif	-	40	3	120	500
Cell15_IMR90_fixed.tif	-	40	3	120	500
Cell16_IMR90_fixed.tif	-	40	3	120	500
Cell17_IMR90_fixed.tif	-	40	3	120	500
Cell18_IMR90_fixed.tif	-	40	3	120	500

Supplementary table 7 (continued)

Name	Cell stage	Time (s)	Exposure time (ms)	Channels*	Frame rate (ms)
n001_G_Aph_4_AT_c01.tif	G1	40	300	1	500
n001_G_Aph_4_AT_c02.tif	G1	-	500	3	-
n001_G_Aph_4_BT_c01.tif	G1	40	300	1	500
n001_G_Aph_4_BT_c02.tif	G1	-	500	3	-
n002_G_Aph_4_AT_c01.tif	G2	40	300	1	500
n002_G_Aph_4_AT_c02.tif	G2	-	500	3	-
n002_G_Aph_4_BT_c01.tif	G2	40	300	1	500
n002_G_Aph_4_BT_c02.tif	G2	-	500	3	-
n003_G_Aph_5_AT_c01.tif	G2	40	300	1	500
n003_G_Aph_5_AT_c02.tif	G2	-	500	3	-
n003_G_Aph_5_BT_c01.tif	G2	40	300	1	500
n003_G_Aph_5_BT_c02.tif	G2	-	500	3	-
n004_G_Aph_5_AT_c01.tif	G2	40	300	1	500
n004_G_Aph_5_AT_c02.tif	G2	-	500	3	-
n004_G_Aph_5_BT_c01.tif	G2	40	300	1	500
n004_G_Aph_5_BT_c02.tif	G2	-	500	3	-

Supplementary table 8

Data description for 2D confocal live images

n005_G_Aph_5_AT_c01.tif	G1	40	300	1	500
n005_G_Aph_5_AT_c02.tif	G1	-	500	3	-
n005_G_Aph_5_BT_c01.tif	G1	40	300	1	500
n005_G_Aph_5_BT_c02.tif	G1	-	500	3	-
n006_G_Aph_6_AT_c01.tif	G2	40	300	1	500
n006_G_Aph_6_AT_c02.tif	G2	-	500	3	-
n006_G_Aph_6_BT_c01.tif	G2	40	300	1	500
n006_G_Aph_6_BT_c02.tif	G2	-	500	3	-
n007_G_Aph_10_AT_c01.tif	G1	40	300	1	500
n007_G_Aph_10_AT_c02.tif	G1	-	500	3	-
n007_G_Aph_10_BT_c01.tif	G1	40	300	1	500
n007_G_Aph_10_BT_c02.tif	G1	-	500	3	-
n008_G_Aph_10_AT_c01.tif	G1	40	300	1	500
n008_G_Aph_10_AT_c02.tif	G1	-	500	3	-
n008_G_Aph_10_BT_c01.tif	G1	40	300	1	500
n008_G_Aph_10_BT_c02.tif	G1	-	500	3	-
n009_G_Aph_10_AT_c01.tif	G1	40	300	1	500
n009_G_Aph_10_AT_c02.tif	G1	-	500	3	-

Supplementary table 8 (continued)

n009_G_Aph_10_BT_c01.tif	G1	40	300	1	500
n009_G_Aph_10_BT_c02.tif	G1	-	500	3	-
n0012_G_Aph_11_AT_c01.tif	G2	40	300	1	500
n0012_G_Aph_11_AT_c02.tif	G2	-	500	3	-
n0012_G_Aph_11_BT_c01.tif	G2	40	300	1	500
n0012_G_Aph_11_BT_c02.tif	G2	-	500	3	-
n0013_G_Aph_11_AT_c01.tif	G2	40	300	1	500
n0013_G_Aph_11_AT_c02.tif	G2	-	500	3	-
n0013_G_Aph_11_BT_c01.tif	G2	40	300	1	500
n0013_G_Aph_11_BT_c02.tif	G2	-	500	3	-
n0015_G_Aph_15_AT_c01.tif	G1	40	300	1	500
n0015_G_Aph_15_AT_c02.tif	G1	-	500	3	-
n0015_G_Aph_15_BT_c01.tif	G1	40	300	1	500
n0015_G_Aph_15_BT_c02.tif	G1	-	500	3	-
n0017_G_Aph_16_AT_c01.tif	G1	40	300	1	500
n0017_G_Aph_16_AT_c02.tif	G1	-	500	3	-
n0017_G_Aph_16_BT_c01.tif	G1	40	300	1	500
n0017_G_Aph_16_BT_c02.tif	G1	-	500	3	-

Supplementary table 8 (continued)

n0018_G_Aph_18_AT_c01.tif	G1	40	300	1	500
n0018_G_Aph_18_AT_c02.tif	G1	-	500	3	-
n0018_G_Aph_18_BT_c01.tif	G1	40	300	1	500
n0018_G_Aph_18_BT_c02.tif	G1	-	500	3	-
n0020_G_Aph_21_AT_c01.tif	G2	40	300	1	500
n0020_G_Aph_21_AT_c02.tif	G2	-	500	3	-
n0020_G_Aph_21_BT_c01.tif	G2	40	300	1	500
n0020_G_Aph_21_BT_c02.tif	G2	-	500	3	-
n001_S_Aph_1_AT_c01.tif	S	40	300	1	500
n001_S_Aph_1_AT_c02.tif	S	-	500	3	-
n001_S_Aph_1_BT_c01.tif	S	40	300	1	500
n001_S_Aph_1_BT_c02.tif	S	-	500	3	-
n002_S_Aph_1_AT_c01.tif	S	40	300	1	500
n002_S_Aph_1_AT_c02.tif	S	-	500	3	-
n002_S_Aph_1_BT_c01.tif	S	40	300	1	500
n002_S_Aph_1_BT_c02.tif	S	-	500	3	-
n004_S_Aph_1_AT_c01.tif	S	40	300	1	500
n004_S_Aph_1_AT_c02.tif	S	-	500	3	-

Supplementary table 8 (continued)

n004_S_Aph_1_BT_c01.tif	S	40	300	1	500
n004_S_Aph_1_BT_c02.tif	S	-	500	3	-
n006_S_Aph_2_AT_c01.tif	S	40	300	1	500
n006_S_Aph_2_AT_c02.tif	S	-	500	3	-
n006_S_Aph_2_BT_c01.tif	S	40	300	1	500
n006_S_Aph_2_BT_c02.tif	S	-	500	3	-
n008_S_Aph_4_AT_c01.tif	S	40	300	1	500
n008_S_Aph_4_AT_c02.tif	S	-	500	3	-
n008_S_Aph_4_BT_c01.tif	S	40	300	1	500
n008_S_Aph_4_BT_c02.tif	S	-	500	3	-
n009_S_Aph_5_AT_c01.tif	S	40	300	1	500
n009_S_Aph_5_AT_c02.tif	S	-	500	3	-
n009_S_Aph_5_BT_c01.tif	S	40	300	1	500
n009_S_Aph_5_BT_c02.tif	S	-	500	3	-
n0012_S_Aph_6_AT_c01.tif	S	40	300	1	500
n0012_S_Aph_6_AT_c02.tif	S	-	500	3	-
n0012_S_Aph_6_BT_c01.tif	S	40	300	1	500
n0012_S_Aph_6_BT_c02.tif	S	-	500	3	-

Supplementary table 8 (continued)

n0013_S_Aph_7_AT_c01.tif	S	40	300	1	500
n0013_S_Aph_7_AT_c02.tif	S	-	500	3	-
n0013_S_Aph_7_BT_c01.tif	S	40	300	1	500
n0013_S_Aph_7_BT_c02.tif	S	-	500	3	-
n0014_S_Aph_7_AT_c01.tif	S	40	300	1	500
n0014_S_Aph_7_AT_c02.tif	S	-	500	3	-
n0014_S_Aph_7_BT_c01.tif	S	40	300	1	500
n0014_S_Aph_7_BT_c02.tif	S	-	500	3	-
n0015_S_Aph_8_AT_c01.tif	S	40	300	1	500
n0015_S_Aph_8_AT_c02.tif	S	-	500	3	-
n0015_S_Aph_8_BT_c01.tif	S	40	300	1	500
n0015_S_Aph_8_BT_c02.tif	S	-	500	3	-
n0016_S_Aph_8_AT_c01.tif	S	40	300	1	500
n0016_S_Aph_8_AT_c02.tif	S	-	500	3	-
n0016_S_Aph_8_BT_c01.tif	S	-	500	3	500
n0016_S_Aph_8_BT_c02.tif	S	40	300	1	-
n0017_S_Aph_8_AT_c01.tif	S	-	500	3	500
n0017_S_Aph_8_AT_c02.tif	S	40	300	1	-

Supplementary table 8 (continued)

n0017_S_Aph_8_BT_c01.tif	S	-	500	3	500
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n0018_S_Aph_8_AT_c01.tif	S	-	500	3	500
n0018_S_Aph_8_AT_c02.tif	S	40	300	1	-
n0018_S_Aph_8_BT_c01.tif	S	-	500	3	500
n0018_S_Aph_8_BT_c02.tif	S	40	300	1	-
n0019_S_Aph_9_AT_c01.tif	S	-	500	3	500
n0019_S_Aph_9_AT_c02.tif	S	40	300	1	-
n0019_S_Aph_9_BT_c01.tif	S	-	500	3	500
n0019_S_Aph_9_BT_c02.tif	S	-	500	3	-
n0020_S_Aph_9_AT_c01.tif	S	40	300	1	500
n0020_S_Aph_9_AT_c02.tif	S	-	500	3	-
n0020_S_Aph_9_BT_c01.tif	S	40	300	1	500
n0020_S_Aph_9_BT_c02.tif	S	-	500	3	-
n0021_S_Aph_1_0_AT_c01.tif	S	40	300	1	500
n0021_S_Aph_1_0_AT_c02.tif	S	-	500	3	-
n0021_S_Aph_1_0_BT_c01.tif	S	40	300	1	500
n0021_S_Aph_1_0_BT_c02.tif	S	-	500	3	-

Supplementary table 8 (continued)

n0022_S_Aph_1_2_AT_c01.tif	S	40	300	1	500
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n0022_S_Aph_1_2_BT_c01.tif	S	40	300	1	500
n0022_S_Aph_1_2_BT_c02.tif	S	-	500	3	-
n0023_S_Aph_1_3_AT_c01.tif	S	40	300	1	500
n0023_S_Aph_1_3_AT_c02.tif	S	-	500	3	-
n0023_S_Aph_1_3_BT_c01.tif	S	40	300	1	500
n0023_S_Aph_1_3_BT_c02.tif	S	-	500	3	-
n0024_S_Aph_1_4_AT_c01.tif	S	40	300	1	500
n0024_S_Aph_1_4_AT_c02.tif	S	-	500	3	-
n0024_S_Aph_1_4_BT_c01.tif	S	40	300	1	500
n0024_S_Aph_1_4_BT_c02.tif	S	-	500	3	-
n0025_S_Aph_1_7_AT_c01.tif	S	40	300	1	500
n0025_S_Aph_1_7_AT_c02.tif	S	-	500	3	-
n0025_S_Aph_1_7_BT_c01.tif	S	40	300	1	500
n0025_S_Aph_1_7_BT_c02.tif	S	-	500	3	-
n0026_S_Aph_1_7_AT_c01.tif	S	40	300	1	500
n0026_S_Aph_1_7_AT_c02.tif	S	-	500	3	-

Supplementary table 8 (continued)

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n0027_S_Aph_21_AT_c02.tif	S	-	500	3	-
n0027_S_Aph_21_BT_c01.tif	S	40	300	1	500
n0027_S_Aph_21_BT_c02.tif	S	-	500	3	-
n0028_S_Aph_21_AT_c01.tif	S	40	300	1	500
n0028_S_Aph_21_AT_c02.tif	S	-	500	3	-
n0028_S_Aph_21_BT_c01.tif	S	40	300	1	500
n0028_S_Aph_21_BT_c02.tif	S	-	500	3	-
n0001_coloc_c02.tif	S	40	300	2	500
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n0003_coloc_c02.tif	S	40	300	2	500
n0004_coloc_c02.tif	S	40	300	2	500
n0005_coloc_c02.tif	S	40	300	2	500
n0006_coloc_c02.tif	S	40	300	2	500
n0007_coloc_c02.tif	S	40	300	2	500
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Supplementary table 8 (continued)

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n0012_coloc_c02.tif	S	40	300	2	500
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n0015_coloc_c02.tif	S	40	300	2	500
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n0017_coloc_c02.tif	S	40	300	2	500
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Supplementary table 8 (continued)

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n002_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n003_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
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n006_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n007_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n008_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n009_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n010_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n011_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500

Supplementary table 8 (continued)

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n013_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n014_G1_IMR90_PCNA_Cy3 dUTP_60x.tif	G1	40	300	2	500
n001_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500
n002_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500
n003_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500
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n007_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500
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n014_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500
n015_G2_IMR90_PCNA_Cy3 dUTP_60x.tif	G2	40	300	2	500

Supplementary table 8 (continued)

n001_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
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n003_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n004_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n005_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
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n007_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n008_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n009_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n010_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n011_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n012_S_IMR90_PCNA_Cy3d UTP_60x.tif	S	40	300	2	500
n0013_S_IMR90_PCNA_Cy3 dUTP_60x.tif	S	40	300	2	500
n0014_S_IMR90_PCNA_Cy3 dUTP_60x.tif	S	40	300	2	500

Supplementary table 8 (continued)

Name	Version	Website	Company/University	Application
Volocity	6.3	-	PerkinElmer, USA	Acquiring live cell time lapses
ImageJ	1.53c	https://imagej.nih.gov/ij/	Wayne Rasband, National Institutes of Health, USA	Image processing and image analysis
RStudio	1.1.447-1.2.5033	https://rstudio.com/	RStudio	Statistical analysis and plotting
Harmony	3.5.1	https://www.perkinelmer.com/product/harmony-4-8-office-hh17000001	PerkinElmer, USA	High content microscopy imaging and analysis
KNIME Analytics Platform	3.5.2	https://www.knime.com/knime-analytics-platform	KNIME AG, Switzerland	High content microscopy image processing and analysis
Adobe Illustrator CS6	16	https://www.adobe.com/	Adobe, USA	Graphical sketch and figures arrangement

Supplementary table 9

Software and macros

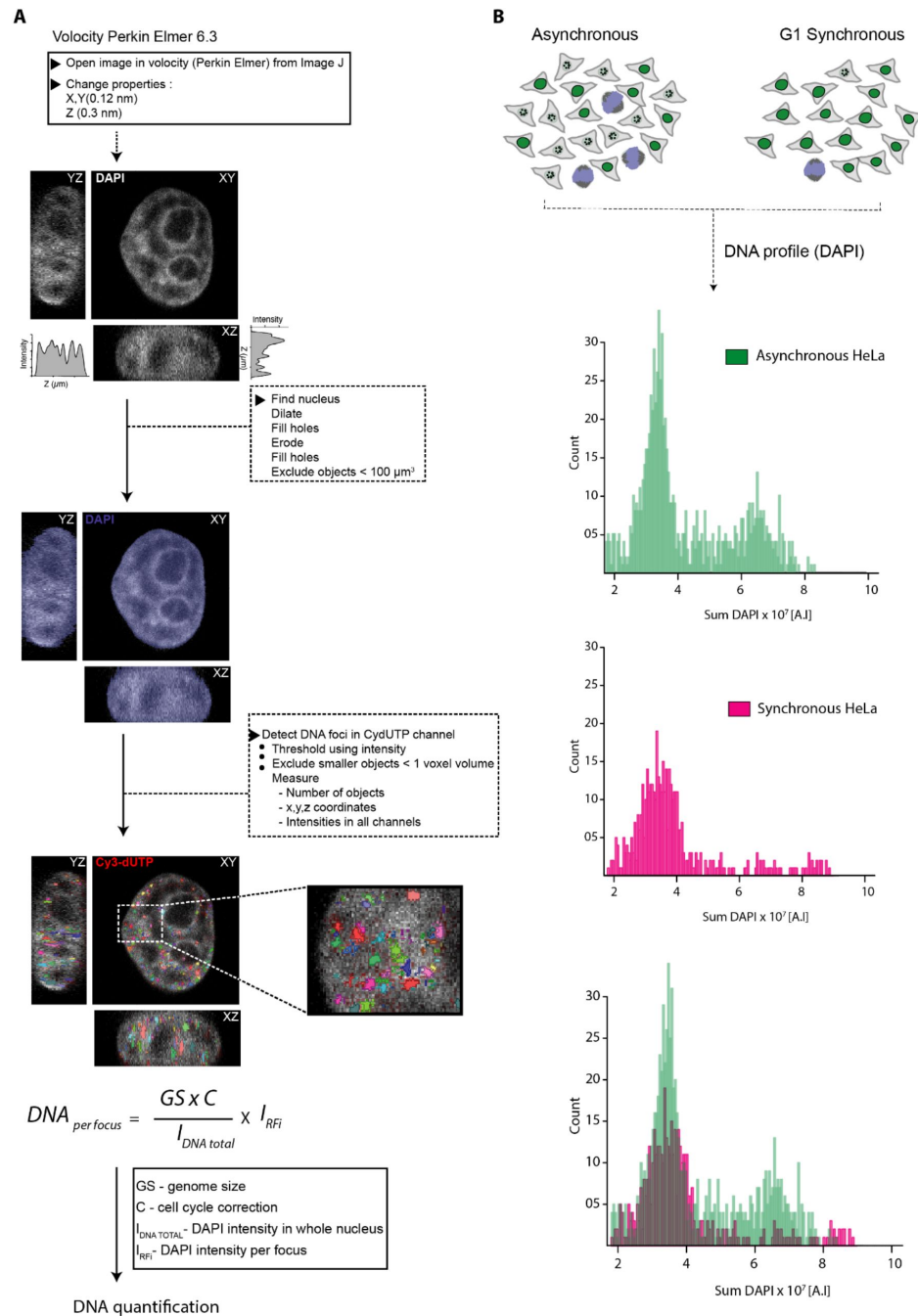


Figure S1

Pipeline and controls for DNA quantification of labeled replication foci using confocal data.

(A) All image sequences were acquired using spinning disk microscopy (60x objective, 1.45 numerical aperture). For analysis, the (16 bit) images were imported from ImageJ into PerkinElmer Velocity 6.3 software. Using the DAPI channel, nuclear segmentation (object recognition) was performed to obtain a mask for the whole nuclear volume and determine the total DNA intensity (mask highlighted with blue). The labeled chromatin foci (Cy3-dUTP) were detected using the intensity and segmented in 3D (highlighted in color palette). The sum DAPI intensity within the segmented foci (I_{RFi}) were also obtained. The ratio of DNA amount per focus and total DNA in the nucleus corrected for cell cycle stage gives the DNA amount per focus. For quantification see [Figure 2](#). The intensity values of the total DAPI intensity within each segmented replication foci (I_{RFi}) are plotted as a histogram. **(B)** HeLa Kyoto cells, either synchronously (magenta) (Methods) or asynchronously growing (green), were seeded on coverslips. The sum DAPI (DNA profile) of cells was plotted individual and in an overlap histogram (brown).

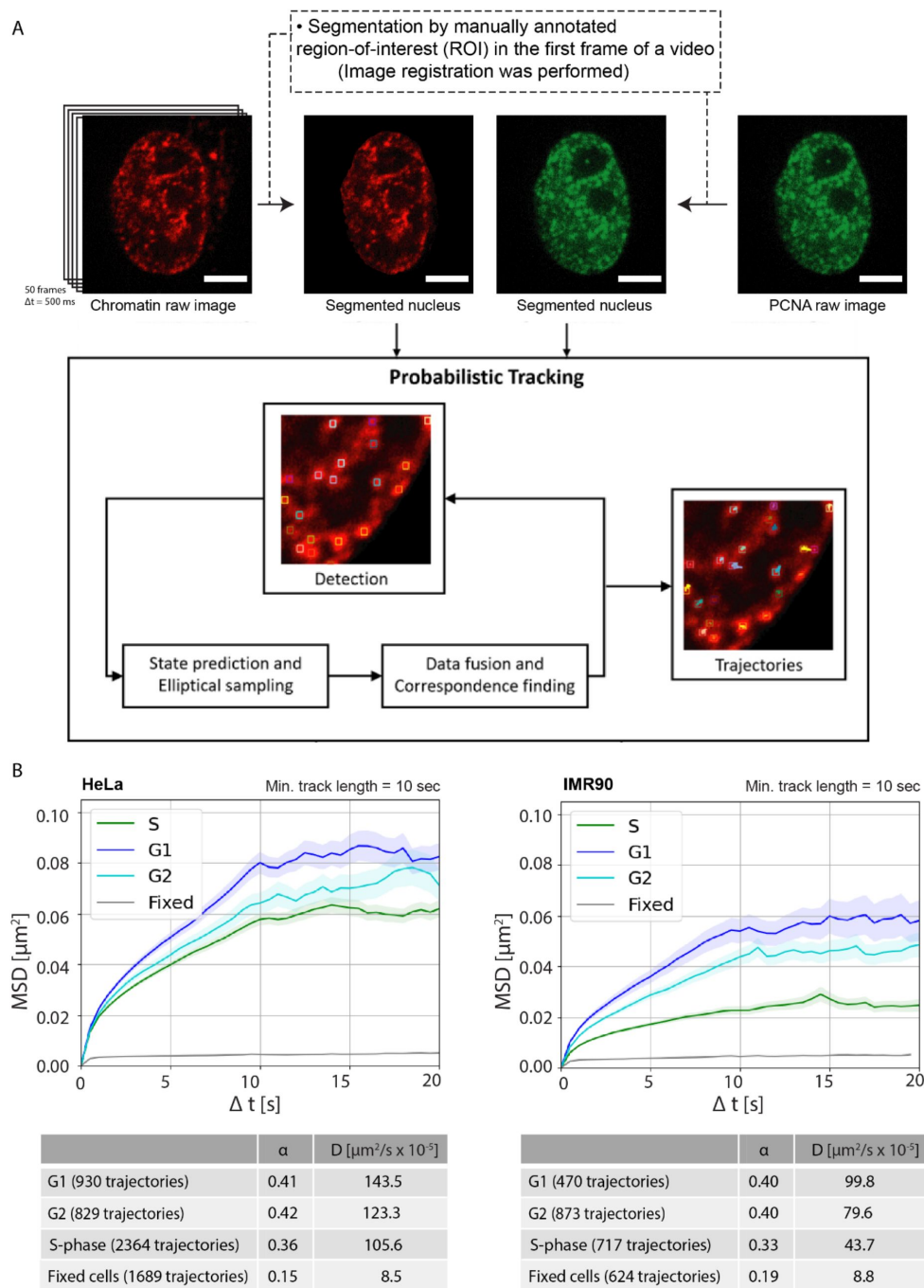


Figure S2

Workflow of chromatin motion and proximity analysis for confocal data.

(A) Raw images of chromatin and PCNA were first segmented to remove the background noise outside the nucleus. In case of IMR90 cells, affine image registration was performed to address the stronger cell movement compared to HeLa cells (Methods). The probabilistic tracking method was then performed which includes detection of chromatin particles and tracking over the time to obtain trajectories. The tracks were then used to plot MSD curves or perform proximity analysis with PCNA. **(B)** Result of motion analysis of computed chromatin tracks for G1, G2, S-phase cell cycle stages (Figure 3) for HeLa and IMR90 cells. Mean Square Displacement (MSD, μm^2) curves were plotted over time (s). Mean square displacement curves for G1/G2, S-phase, fixed cells with a minimum track length of 10 seconds and a total time of 20 seconds were plotted with error bars (SEM-standard error of the mean) representing the deviations between the MSD curves for an image sequence in transparent color around the curve. Scale bar: 10 μm .

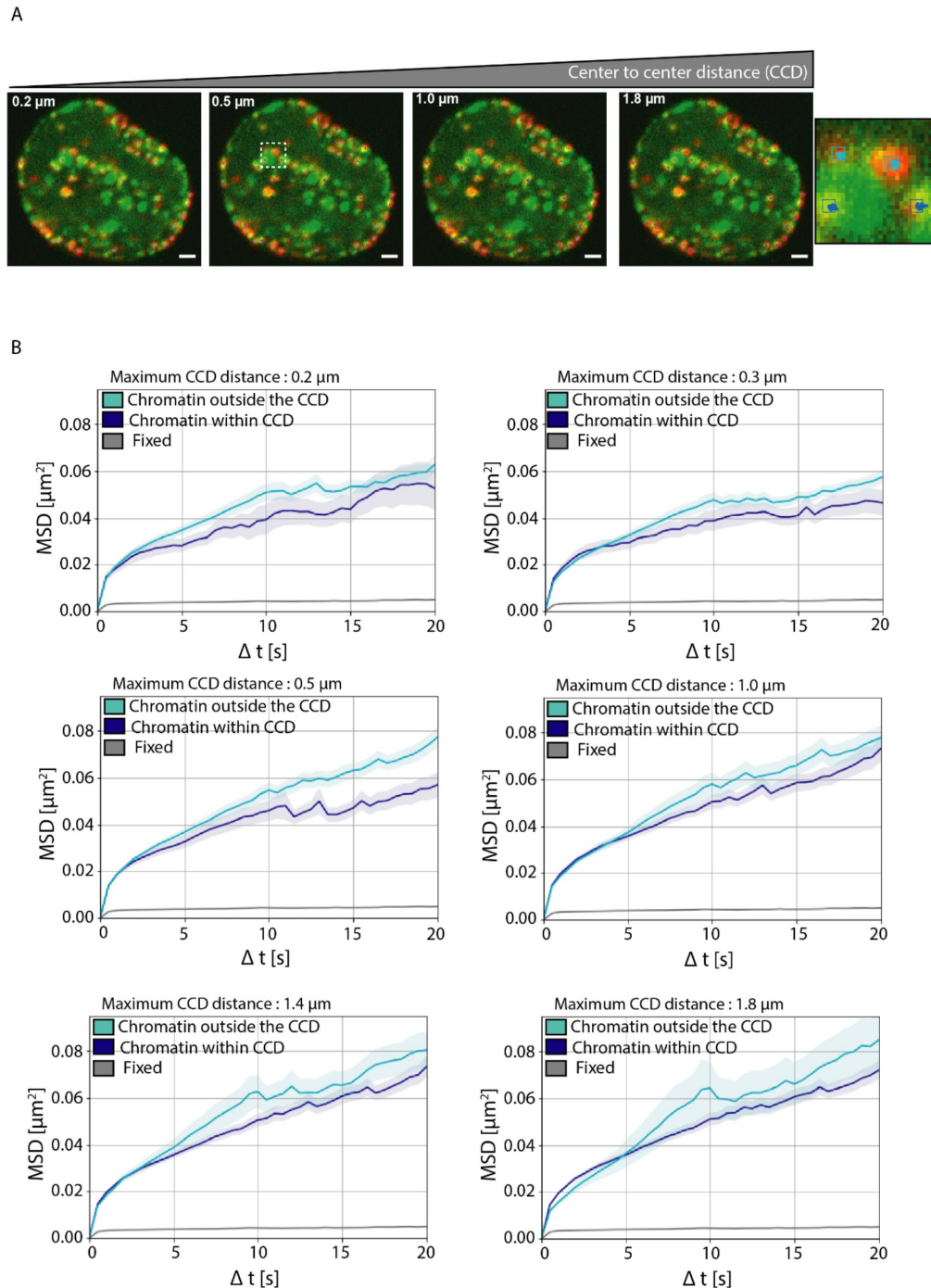


Figure S3

Mean square displacement analysis of labeled chromatin in the proximity of active replication sites at varying center to center distances.

(A) Overlay images of GFP-PCNA (active replication sites, green) and labeled chromatin (red). Chromatin outside the center to center distance (CCD) is highlighted with cyan dots and chromatin within center to center distance (CCD) is highlighted with dark blue dots within the magnified inset (demarcated by a white box). **(B)** Mean square displacement (MSD) curves of chromatin outside the center to center distance (CCD) and chromatin within center to center distance (CCD) along with fixed cell control with increasing distance from 0.2 μm to 1.8 μm . This allows us to analyze the effect of replication factors on chromatin mobility at increasing distances. The MSD curves were plotted with error bars (standard deviation) represented in transparent color around the curve. Scale bar: 1 μm .

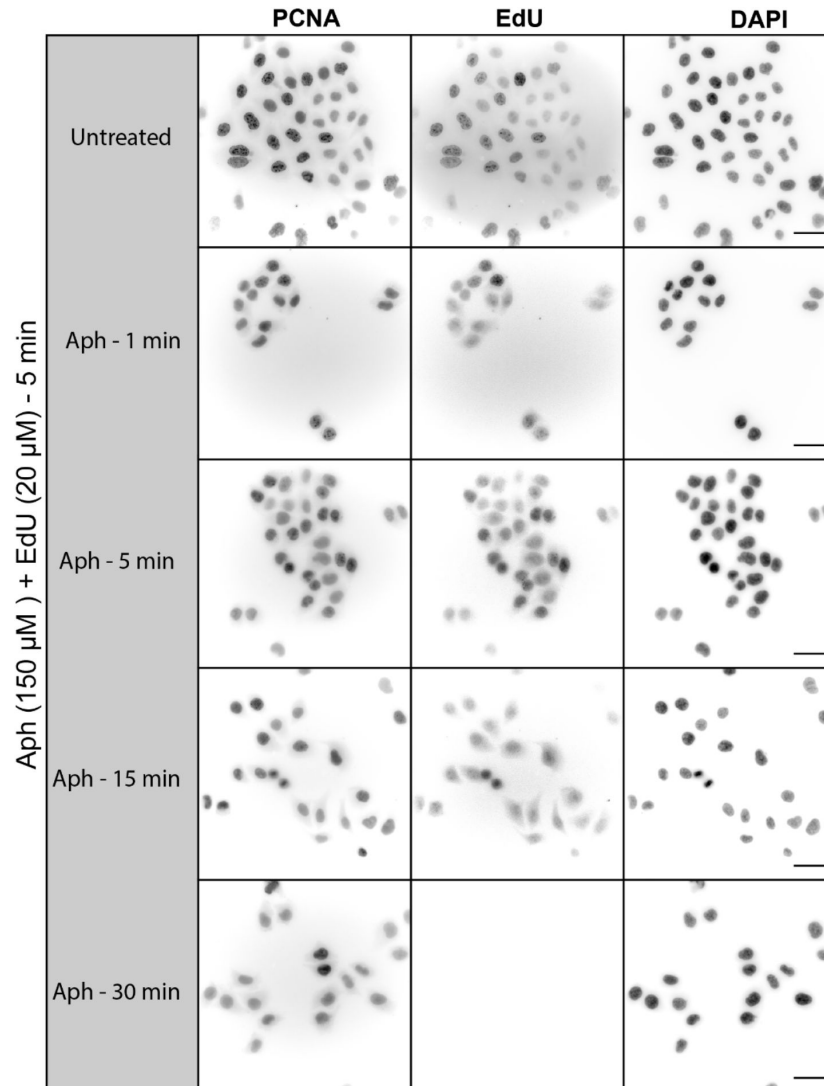


Figure S4

Time course analysis of inhibition of DNA synthesis upon aphidicolin treatment.

(A) HeLa cells stably expressing GFP-PCNA were treated with aphidicolin (150 μ M). Time course of treatments depicting the distribution of PCNA as well as the incorporation of the thymidine analog EdU is shown. EdU was used to detect DNA synthesis. For quantification see **Figure 5A**. Scale bar: 100 μ m.

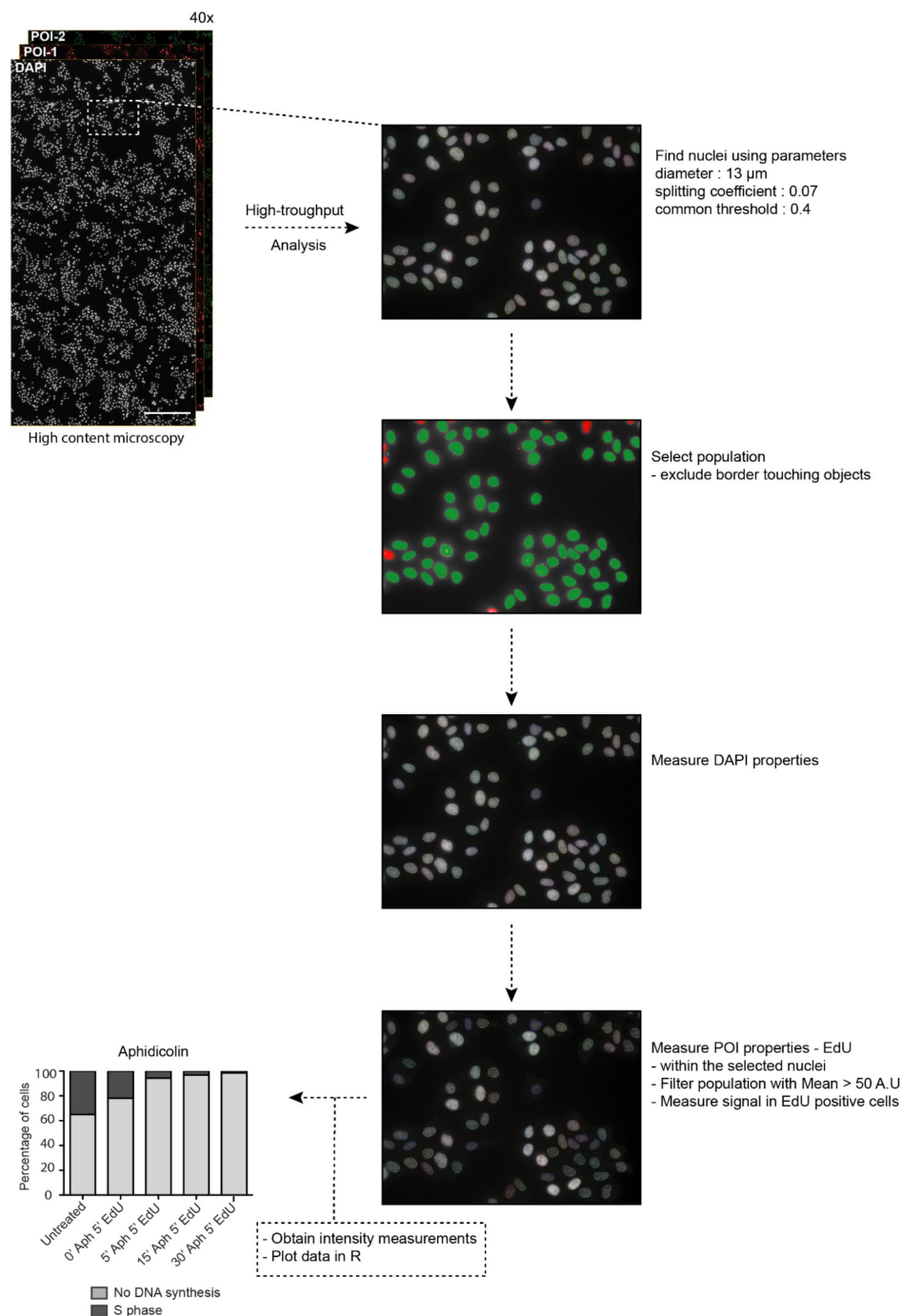


Figure S5

Schematic overview of the high throughput analysis pipeline of DNA synthesis inhibition by aphidicolin.

Images were obtained using the 40x objective (numerical aperture 0.95) in the DAPI and EdU channels. Once the images were acquired they were analyzed using the Harmony software from PerkinElmer. Briefly, the DAPI channel was used to segment nuclei using properties including diameter, splitting coefficient and threshold. Once the nuclei were segmented, the population cells were then selected for analysis by removing the objects either touching the border or fused. In the population of cells selected, the properties including mean, median, standard deviation and sum intensities were measured in the DAPI and EdU channels. The data measurements were then imported to R and visualized for the EdU positive population. Scale bar: 500 μ m.

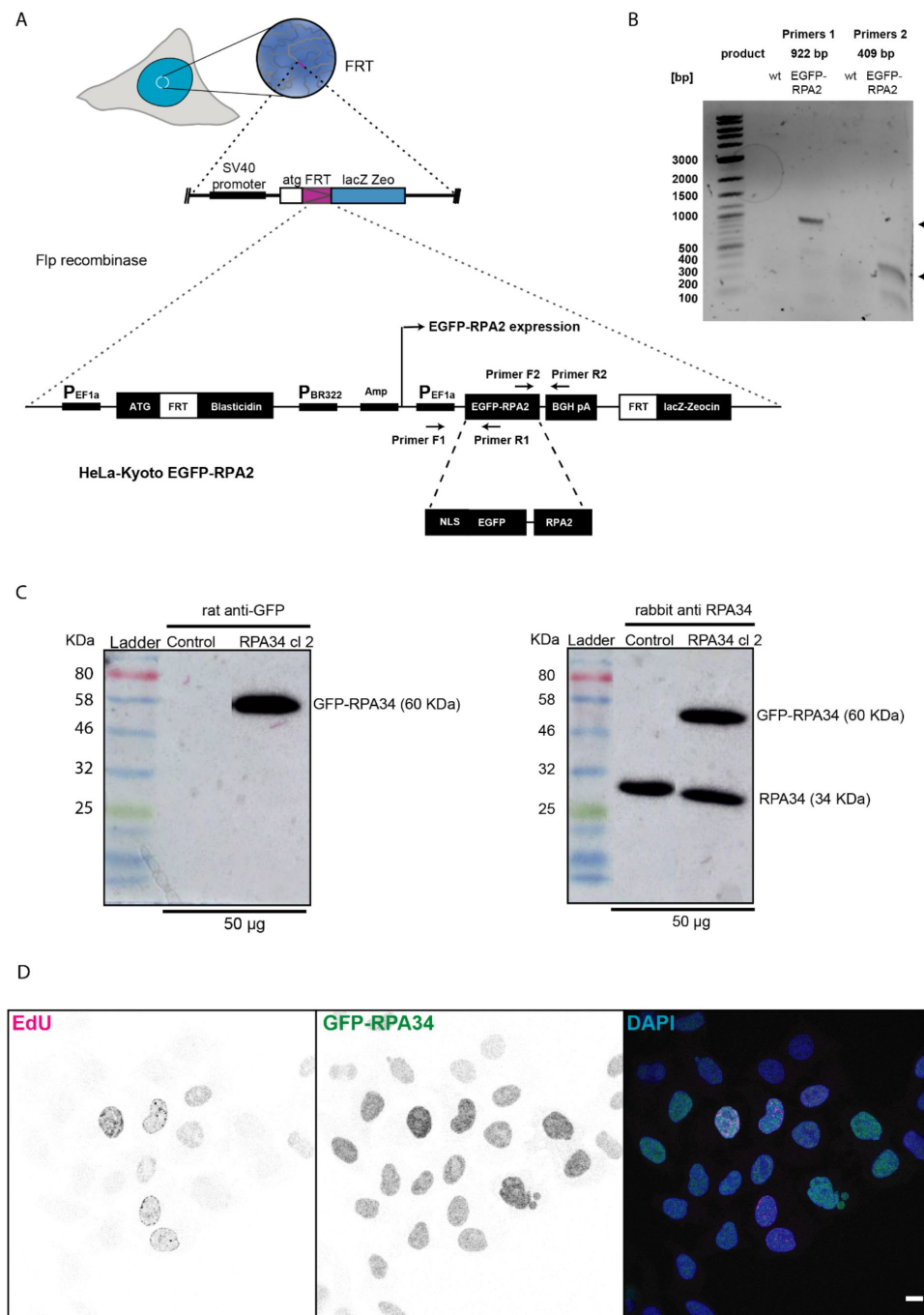


Figure S6

HeLa GFP-RPA cell line characterization after generation using the Flp-recombinase system.

(A) Schematic representation of the insertion of fluorescently tagged RPA34 (aka RPA2) encoding plasmid DNA using the chromosomally inserted FRT site in the previously generated HeLa Kyoto FRTLacZ (Methods). Positive clones were selected using 1 μ g/ml blasticidin in media. **(B)** PCR was performed on genomic DNA isolated from wild type and GFP-RPA34 clones using different primer pairs as indicated. **(C)** Cells from the positive clones were collected to perform a western blot using the anti-GFP antibody to test the level of GFP-RPA34 and an anti-RPA34 antibody was used to detect and compare both endogenous and fusion protein levels in control versus the positive clone. **(D)** The cells were seeded on glass coverslips, fixed and imaged to validate the protein localization relative to active DNA replication sites labeled EdU as well as DNA labeled with DAPI. Scale bar: 5 μ m.

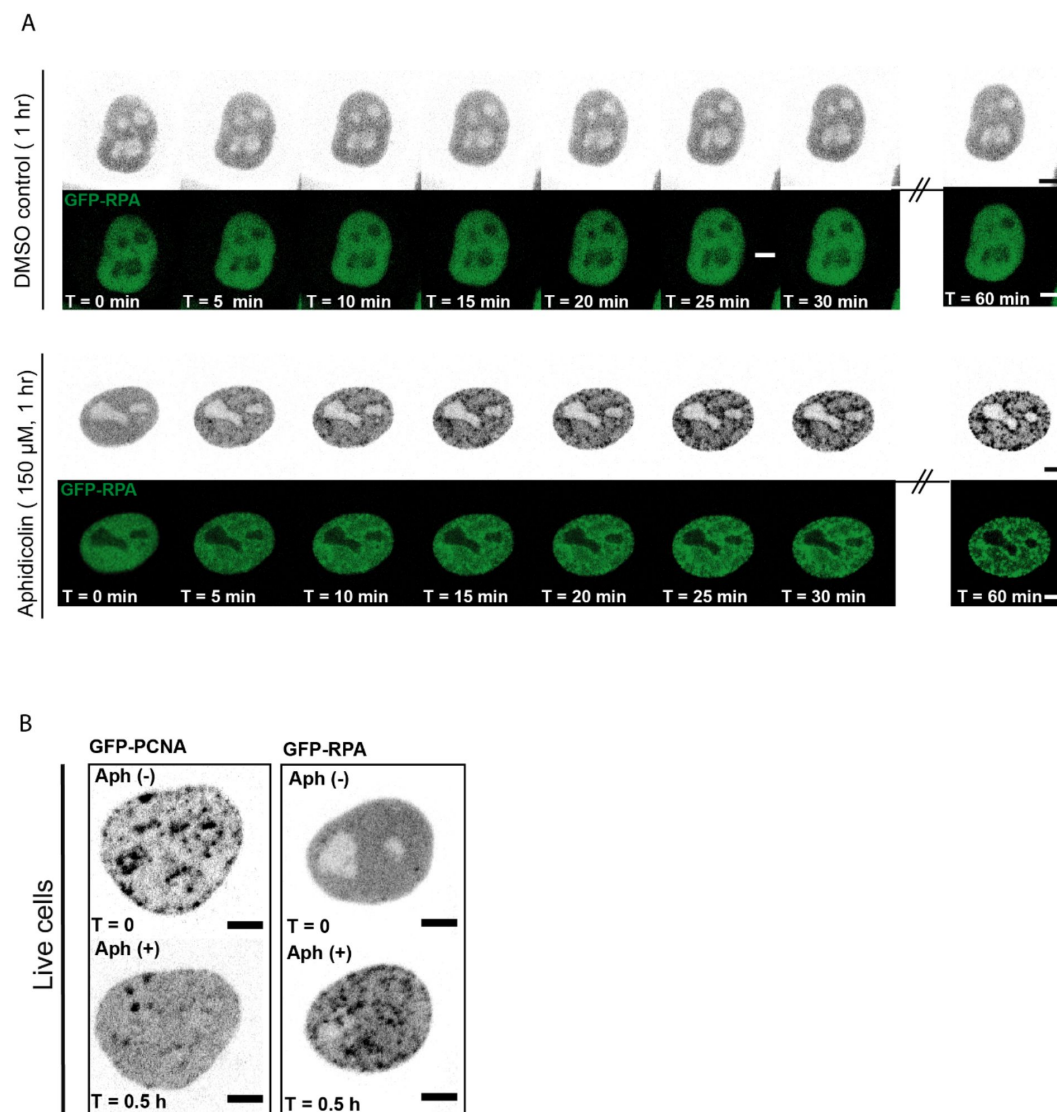


Figure S7

Live-cell time-lapse microscopy of HeLa GFP-RPA34 cells to determine RPA accumulation at replication sites upon aphidicolin treatment.

(A) Confocal time-lapse microscopy of HeLa cells stably expressing GFP-RPA34 treated with DMSO (control), and aphidicolin. Movies were acquired at a frame rate of 5 minutes and at every time a Z stack was acquired. A single Z-plane at different timepoints was shown as a representative image for all three conditions. The time-lapse images were imported to PerkinElmer Velocity 6.3 for analysis and the accumulation of GFP-RPA34 at replication foci was measured over time. **(B)** Representative images of live HeLa cells stably expressing GFP-PCNA or GFP-RPA34 are shown in gray scale before and after aphidicolin treatment. Scale bar: 5 μ m.

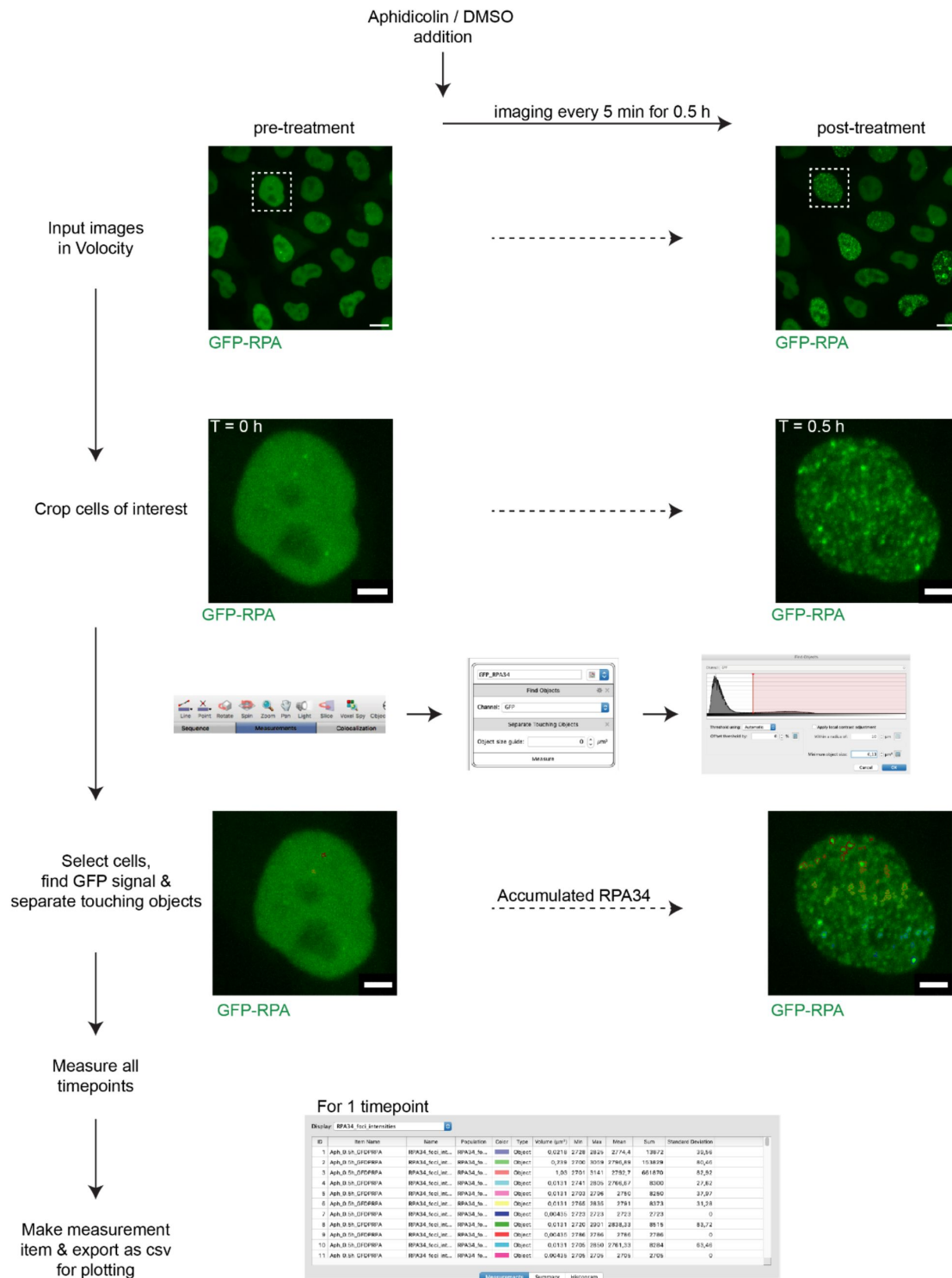


Figure S8

Pipeline for the analysis of RPA enrichment at replication sites in living cells.

The enrichment or accumulation analysis was performed using PerkinElmer Volocity 6.3 software. Briefly cells of interest (white inserts) were cropped and Z-stacks along with different timepoints were imported together. The GFP-RPA34 intensities at different timepoints for aphidicolin (every 5 min / 0.5 h) were measured and the coefficient of variation c_v was calculated (Methods, Accumulation analysis). The intensity values were exported as CSV files for further analysis. The plots were generated in R studio. Scale bar: 5 µm. Cropped cells scale bar

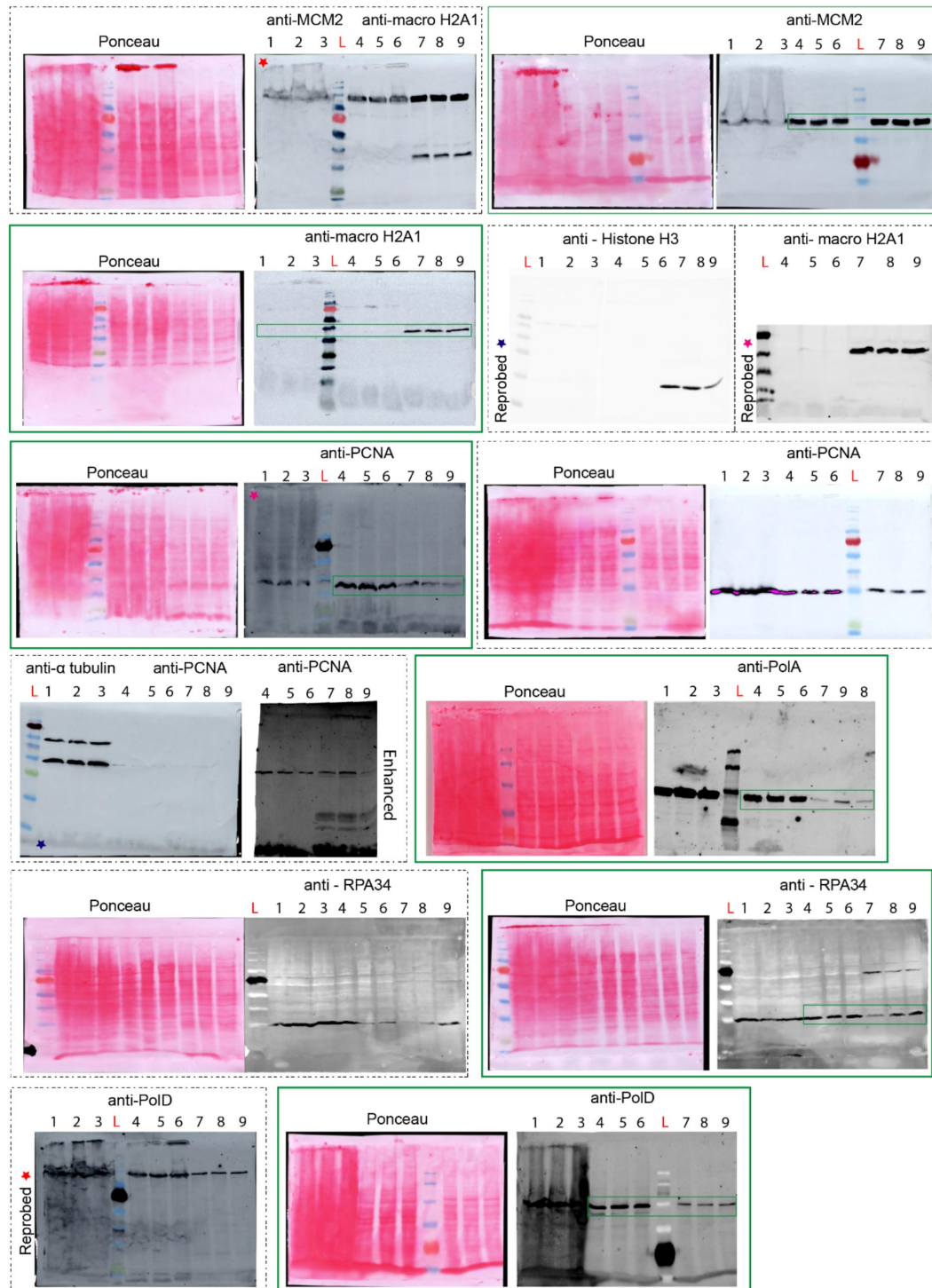


Figure S9

Full length western blots probed for different replication factors with cytoplasm, nucleoplasm and chromatin fractions.

The blots that are reprobed with a different antibody are highlighted with color stars. The blots highlighted with green inserts are the full length and single contrast blots that are cropped for easier visualization as the loading order of samples and ladder is different and are summarized in [Figure 5C](#).

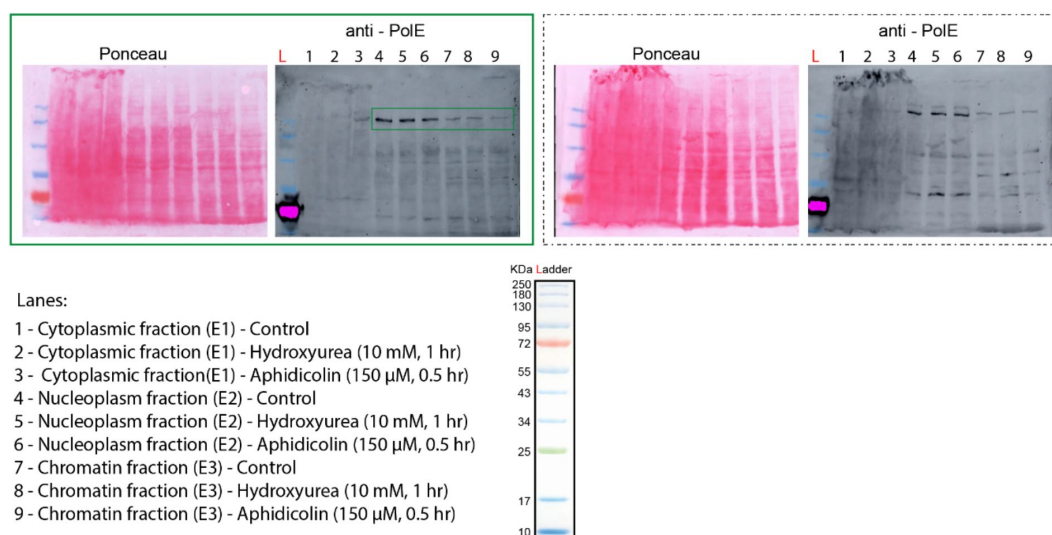


Figure S9 (continued)

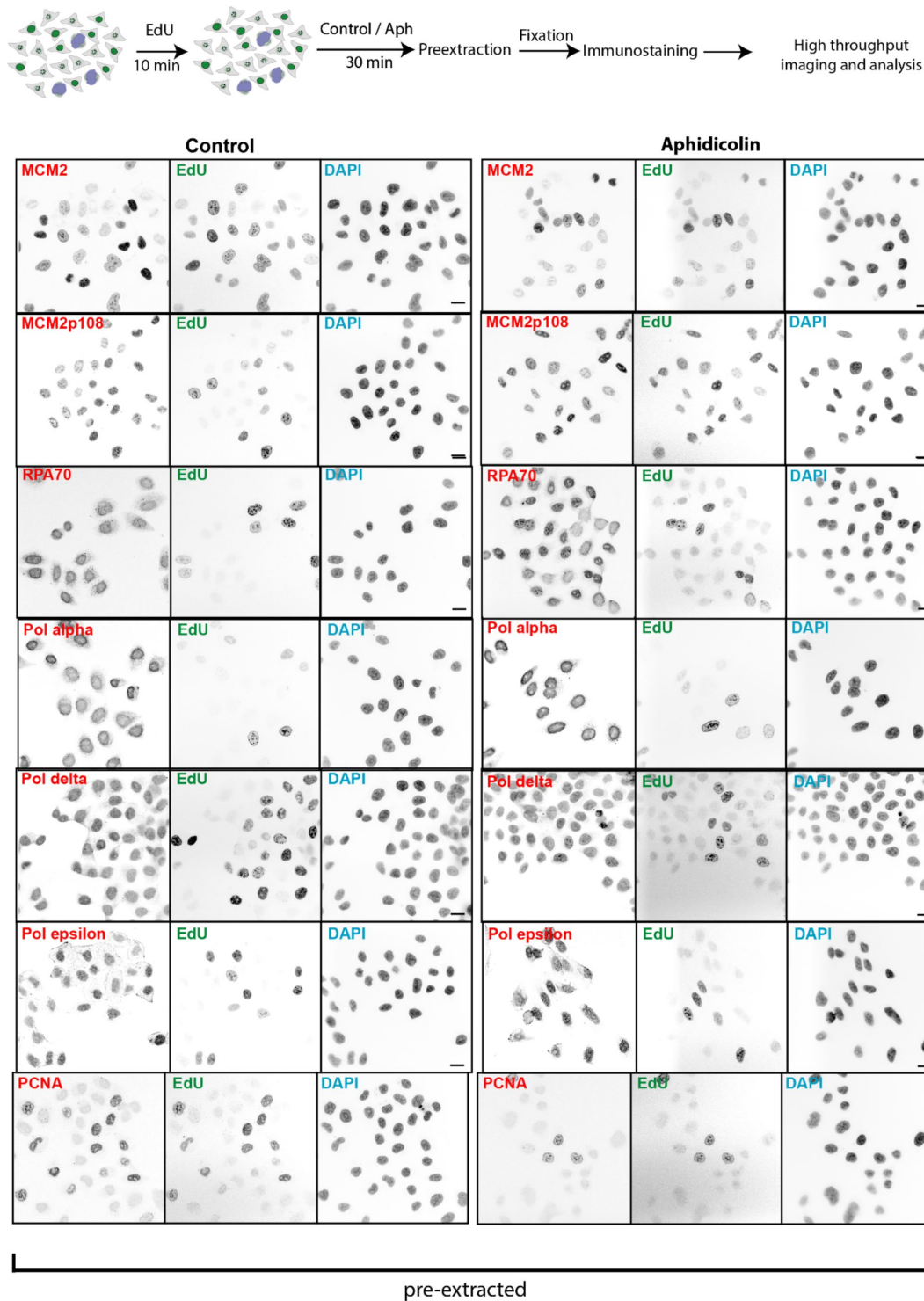


Figure S10

Analysis of replisome component enrichment at replication sites upon replication stress.

Spatial localization and accumulation analysis of DNA replication factors such as DNA helicase subunit MCM2 and active MCM2 phosphorylated at S108, polymerase clamp protein (PCNA), single stranded binding protein (RPA) and DNA polymerases (α , δ , ϵ) catalytic subunits using immunostainings. The cells in S-phase were identified using an EdU pulse before the treatment. Representative low magnification images of protein localization in control and aphidicolin treated HeLa cells. For quantification see [Figure 5D](#). Scale bar: 10 μ m.

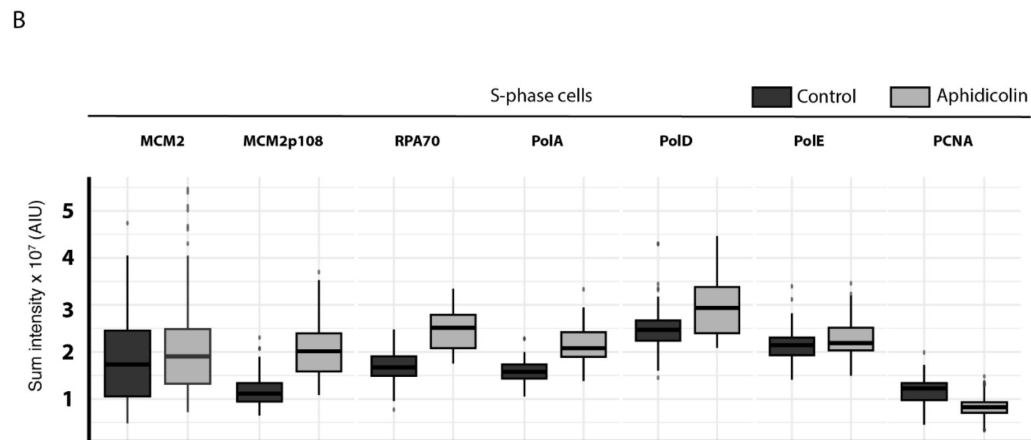
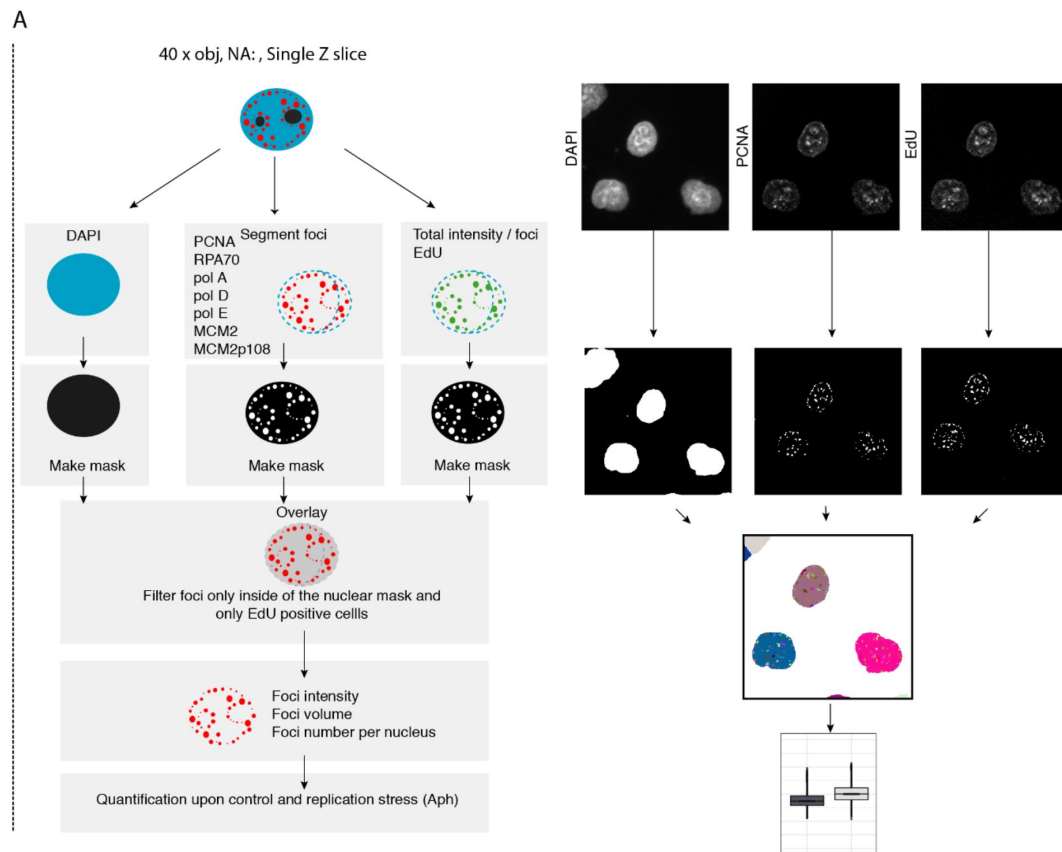


Figure S11

Pipeline for the analysis of replisome components on chromatin upon stress.

(A) The accumulation analysis was performed. The channels were separated, the DAPI channel was used for creation of nuclear mask, the EdU and replication proteins channels were used for replication foci segmentation. The nuclear mask was overlaid with the replication foci mask, so that only foci inside of the nuclear area were selected. The replication foci parameters were measured as follows: focus intensity parameters, focus area, foci number per nucleus (Methods, High throughput image analysis of replisome component). The replication foci values per nucleus together with the DAPI nuclear intensity were exported for further analysis. The plots were generated in R studio. **(B)** Box plots from **Figure 5D** are shown here at the same Y axis scale.

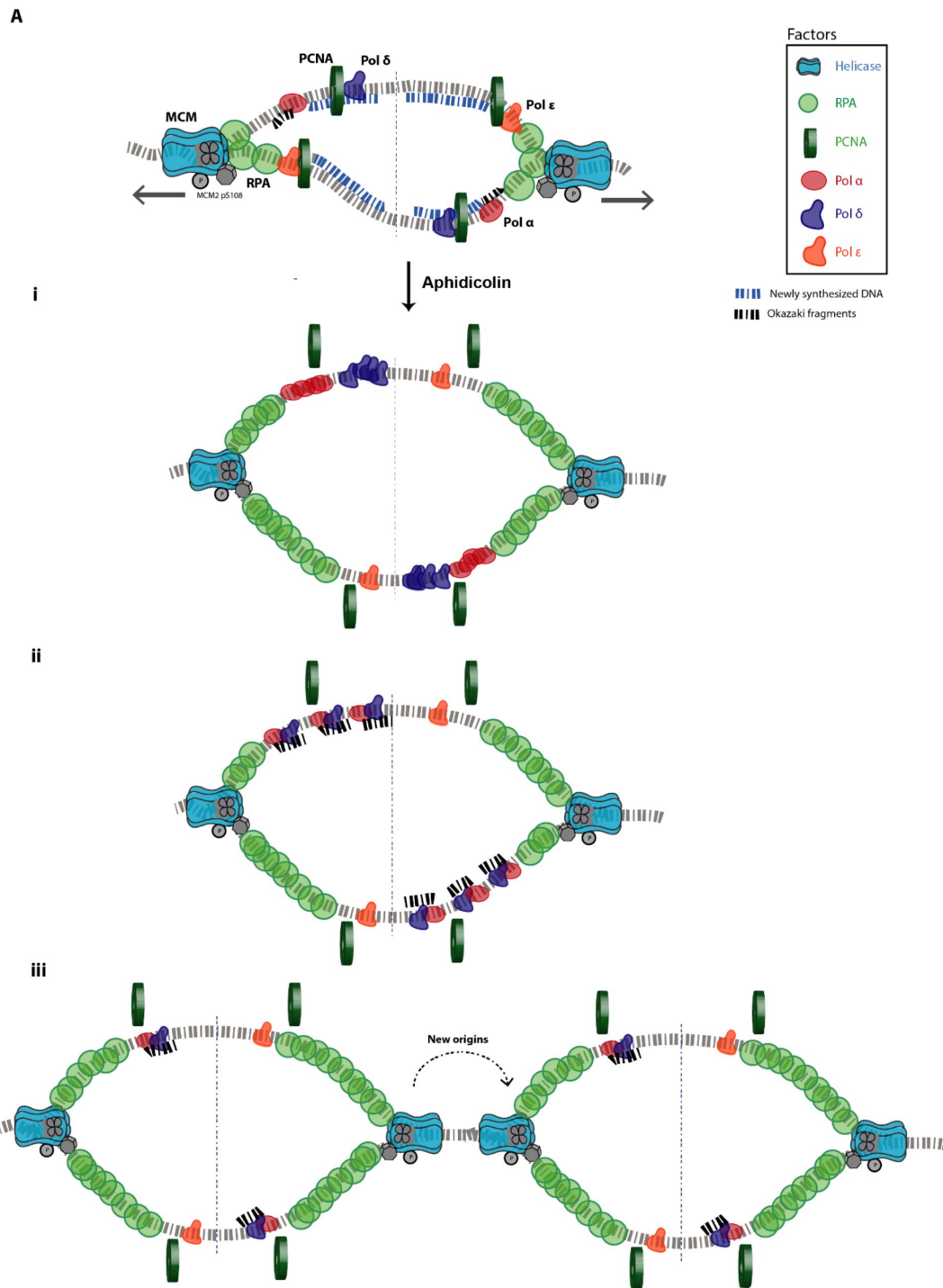


Figure S12

Model describing the different scenarios of replisome response to stress.

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Joint Public Review:

The manuscript presented by Pabba et al. studied chromatin dynamics throughout the cell cycle. The authors used fluorescence signals and patterns of GFP-PCNA (GFP tagged proliferating cell nuclear antigen) and CY3-dUTP (which labels newly synthesized DNA but not the DNA template) to determine cell cycle stages in asynchronized HeLa (Kyoto) cells and

track movements of chromatin domains. PCNA binds to replication forks and form replication foci during the S phase. The major conclusions are: (1) Labeled chromatin domains were more mobile in G1/G2 relative to the S-phase. (2) Restricted chromatin motion occurred at sites in proximity to DNA replication sites. (3) Chromatin motion was restricted by the loading of replisomes, independent of DNA synthesis. This work is based on previous work published in 2015, entitled "4D Visualization of replication foci in mammalian cells corresponding to individual replicons," in which the labeling method was demonstrated to be sound.

Comments on latest version: The revised manuscript has included data from a diploid cell line IMR90 (fibroblasts isolated from normal lung tissue) to strengthen the conclusions. Overall, quality of the work is substantially improved.

Author Response

The following is the authors' response to the original reviews.

eLife assessment

This is a valuable investigation of the chromatin dynamics throughout the cell cycle by using fluorescence signals and patterns of GFP-PCNA and CY3-dUTP, which labels newly synthesized DNA. The authors report reduced chromatin mobility in S relative to G1 phase. The technology and methods used are solid, but the significance of the work is reduced by the model system employed, the HeLa cell line, which has a greatly abnormal genome.

We have obtained data from a diploid human cell that validates the reduction of S-phase chromatin mobility.

Public Review:

The manuscript presented by Pabba et al. studied chromatin dynamics throughout the cell cycle. The authors used fluorescence signals and patterns of GFP-PCNA (GFP tagged proliferating cell nuclear antigen) and CY3-dUTP (which labels newly synthesized DNA but not the DNA template) to determine cell cycle stages in asynchronized HeLa (Kyoto) cells and track movements of chromatin domains. PCNA binds to replication forks and form replication foci during the S phase. The major conclusions are: (1) Labeled chromatin domains were more mobile in G1/G2 relative to the S-phase. (2) Restricted chromatin motion occurred at sites in proximity to DNA replication sites. (3) Chromatin motion was restricted by the loading of replisomes, independent of DNA synthesis. This work is based on previous work published in 2015, entitled "4D Visualization of replication foci in mammalian cells corresponding to individual replicons," in which the labeling method was demonstrated to be sound. Although interesting, reduced chromatin mobility in S relative to G1 phase is not new to the field.

It was first shown in yeast (Heun et al. 2001; DOI:10.1126/science.1065366) that the S-phase mobility is reduced compared to the G1 phase. This was followed by other papers showing the same in yeast [(Gasser 2002; DOI: 10.1126/science.1067703), (Smith et al. 2019; DOI: 10.1091/mbc.E19-08-0469)]. The relation between chromatin motion and cell cycle progression in the mammalian genome is less studied. Over recent years there have been a few studies that addressed chromatin mobility and cell cycle progression but from a different

perspective. In the publication Nozaki et al. (2017; DOI:10.1016/j.molcel.2017.06.018) chromatin motion analysis was performed on single histones. The study did not find a significant change of histone/nucleosome mobility measured during cell cycle progression. Using CRISPR/dCas9 to label random DNA loci, Ma et al. (2019; DOI:10.1083/jcb.201807162) found that chromatin motion in S-phase was significantly lower than in the G1 phase. However, most of the studies measure the chromatin motion using either insertion of ectopic loci or proteins marking the loci (dCas9) or histones. Using either ectopic loci addition or CRISPR/dCas9 might have an effect on the chromatin mobility itself and measuring single histone motion is not equivalent to measuring the motion of DNA segments. We, therefore, opted to label the DNA directly using the replication of the DNA. In this manner we preserve the native chromatin structure and, thus, motion.

Importantly, in addition to measuring decreased DNA motion in S-phase, our study indicates that it is not the DNA synthesis per se but the loading of replisomes onto chromatin that slows down its motion. This allowed us to propose a mechanism on how chromatin motion is affected by DNA replication in S-phase.

The genome in HeLa cells is greatly abnormal with heterogeneous aneuploidy, which makes quantification complicated and weakens the conclusions.

We agree that the HeLa cells are aneuploid and we have addressed the heterogeneity of HeLa Kyoto within our detection methods (for clarification see point 3). To validate our conclusions in normal diploid human cells, we performed the chromatin mobility analysis using human fibroblasts (IMR90 cells in figures 2, 3 and S2) and plotted the MSD curves for different cell cycle stages. The outcome of this analysis showed that the mobility of chromatin in diploid fibroblasts in S-phase is lower than in G1 and G2. In fact, this effect is stronger in IMR90 cells than in HeLa Kyoto cells. Hence, this is not an aneuploid tumor cell phenomenon.

The manuscript is difficult to follow in places due to insufficient clarity. The manuscript should be written in a way that can be understood without referencing previous articles. Overall, the work is moderately impactful to the field.

Major recommendations:

- 1. In Figure 1B, the illustration and images for S phase are confusing. The author should specify which is early S and which is late S. Do the yellow circles represent GFP-PCNA foci? How did the authors distinguish mid S from early S and late S (in Figure 2)? Are all images in Figure 1 scaled to the same contrast threshold?*

The yellow circles correspond to the colocalized signal of GFP-PCNA and Cy3-dUTP that overlap and represent the labeled chromatin sites that are replicated in the next cell cycle.

We clarified all the points mentioned above and updated figure 1 and figure 2 accordingly.

- 1. In Figure 2B, the y-axis is marked as "Frequency of cells" but the equation listed below is counting DNA (per focus). How to convert DNA (per focus) to DNA (per cell)? The x-axis is marked as "Genome size" without any unit (e.g., kb? Mb?) The x-axis seems to be the C factor, not the genome size.*

To determine the amount of DNA present in each labeled DNA focus, we first segmented the whole nucleus and measured the total intensity of DAPI (DNA amount) which is called IDNA TOTAL. Then the labeled replication foci are segmented and the intensity of label present in each segmented foci is measured (IRFi). Throughout the S-phase progression the amount of DNA increases twofold from early to late S-phase. The cells at each cell cycle stage were

determined using the PCNA pattern. By plotting the frequency (number of cells) and the relative genome content normalized to the G1 stage we calculated the relative genome size otherwise called cell cycle correction factor for each stage from G1 to G2. The ratio of DNA intensity in labeled replication (IRFi)/ to the total DNA intensity of DAPI (IDNA total) gives the fraction of DNA present in each foci compared to the whole nucleus. This ratio was then multiplied by the genome size (Kbp) of HeLa Kyoto cells which was measured and published in Chagin et al. (2016; DOI:10.1038/ncomms11231). This gives us the approximate amount of DNA present in each labeled replication foci in Kbp. Since the genome duplicates over cell cycle stages, the measured DNA content in IRFi was corrected to the cell cycle stage (determined by PCNA) by multiplying the cell cycle correction factor.

1. *HeLa cells are known to be highly heterogeneous and heavily aneuploidy. Cells in one sample have different numbers of chromosomes ranging from 50 - 80. Therefore, GS (genome size) for each cell should not be the same. Using one constant GS in the equation for every cell introduces errors. Has the cell-to-cell variation been considered and corrected in the data? If not, the authors should provide information regarding cell-to-cell variations, such as the intensity variation of nuclear DAPI signals in synchronized cells.*

It is true that the HeLa genome is aneuploid. However, the heterogeneity of the genome is true, if one compares different HeLa strains as studied in Frattini et al. (2015; DOI:10.1038/srep15377), where they show the variability of genome and RNA expression profiles and small genomic rearrangements among different HeLa strains. However, to our knowledge, it is not studied extensively or shown whether the heterogeneity and aneuploidy would also be a cell to cell variation. Therefore, we performed a control experiment to verify the variability between HeLa Kyoto cells, where we either synchronized or not and stained with DAPI and the DNA content profiles of all cells were plotted as a histogram (supplementary figure 1B) to show that cell to cell variations is not present and by synchronizing, we see that the cell population in G1, has similar DNA content showing that the cell to cell variability is negligible in our detection methods. Nonetheless, we have obtained data using normal diploid human fibroblasts, which validated our outcome.

STABLE:

Macville, Merry, et al. "Comprehensive and definitive molecular cytogenetic characterization of HeLa cells by spectral karyotyping." *Cancer research* 59.1 (1999): 141-150.

UNSTABLE:

Liu, Yansheng, et al. "Multi-omic measurements of heterogeneity in HeLa cells across laboratories." *Nature biotechnology* 37.3 (2019): 314-322.

Landry, Jonathan JM, et al. "The genomic and transcriptomic landscape of a HeLa cell line." *G3: Genes, Genomes, Genetics* 3.8 (2013): 1213-1224.

1. *The chromatin foci are in a variety of sizes and intensities. How were boundaries of foci determined? Weak foci were picked up in one image but not in another. This is a concern because the size of the chromatin domain could influence mobility measurement. The authors should provide control experiments or better explanations for detecting and selecting chromatin foci.*

The method for detecting chromatin foci is described in "Materials and Methods" section "Automated tracking of chromatin structures in time-lapse videos". "Chromatin structures are detected by the spot-enhancing filter (SEF) (Sage et al., 2005; doi:10.1109/TIP.2005.852787) which consists of a Laplacian-of-Gaussian (LoG) filter followed by thresholding the filtered

image and determination of local maxima. The threshold is automatically determined by the mean of the absolute values of the filtered image plus a factor times the standard deviation.” For reasons of consistency, we used the same threshold factor for all images of an image sequence. Therefore, depending on the intensity distribution in an image, it can happen that weak foci are not detected in some images. Alternatively, one could manually adapt the threshold factor for all single images, which, however, would be subjective. We now added the information that we used the same threshold factor for all images of an image sequence.

1. In Figure 3, the authors combined MSD from G1 and G2 in one group. Has any published data suggested that chromatin dynamics are the same in G1 and G2?

To clarify this we separated G1 and G2 mobility measurements in supplementary figure S2 and updated the figures and text accordingly.

1. In Figure 3B, cytoplasmic CY3-dUTP foci are found in the G1/G2 and S images. Are these CY3-dUTP aggregates? If so, are they also found in the nucleus? What is the mobility of the cytoplasmic CY3-dUTP foci?

These are aggregates and not found in the nucleus. These foci were excluded from the analysis by using a nuclear mask based on the PCNA signal. This information was added to the figure 3B legend.

1. In Figure 4, how is colocalization defined? 1.8 μm is approximately the size of a chromosome territory, which is much larger than 0.5 Mb. Two foci that are 1.8 μm apart should not be considered in the same chromosome.

We agree that colocalized would indeed mean that the signals are overlapping. Therefore, we updated the figures and text as center to center distance or proximity analysis.

Minor comments:

1. Figure 3D should be presented by a box and whisker plot. The histogram does not show an actual distribution of the data.

The histograms shown in figure 3D is the average mean square displacement measurement value for different cell cycle stages. These are the same data shown in the table. Therefore, the histogram is removed and the table in figure 3C is retained.

1. Please explain Figure 3C error bars in the figure legend. Are they SD?

The error bars of the MSD curves (highlighted in bright color around the curves) in figure 3C show the standard error of the mean (SEM) representing the deviations between the MSD curves for an image sequence. We clarified this in the legend of Figure 3C.

1. In Figure 5C, some western blotting results seem to be assembled from replicate experiments. Comparing signals from one experiment with the same background is suggested.

We made sure that the western blots from the same replicates are cropped and the information is also added to the respective figure legends.